



Xi'an Jiaotong-Liverpool University

西交利物浦大学

INT305 Machine Learning

Lecture 12

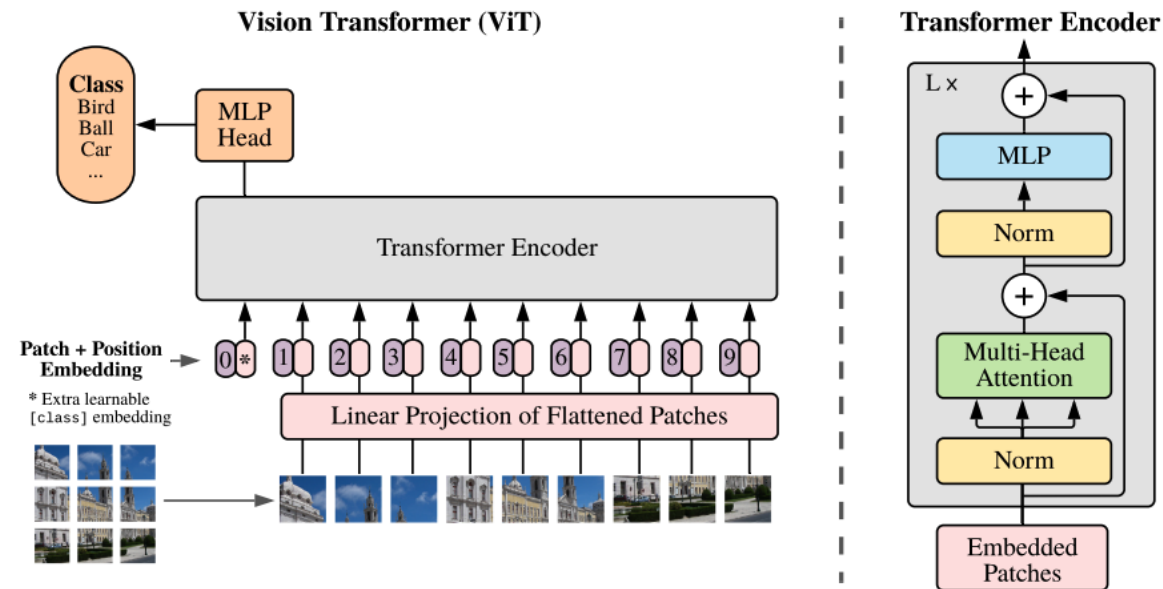
Basic information of MLLMs and Revision

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Basic knowledge of MLLMs- attention mechanism and FFN



From paper: AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

Revision

Supervised Learning, Linear Methods for Regression, Optimization

- First learning algorithm: **KNN**
- Second learning algorithm of the course: **linear regression**.
 - **Task**: predict scalar-valued targets (e.g. stock prices)
 - **Architecture**: linear function of the inputs
- While **KNN** was a complete algorithm, linear regression exemplifies a modular approach that will be used through this course:
 - choose a **model** describing the relationships between variables of interest
 - define a **loss function** quantifying how bad the fit to the data is
 - choose a **regularizer** saying how much we prefer different candidate models (or explanations of data)
 - fit a model that minimizes the loss function and satisfies the constraint/penalty imposed by the regularizer, possibly using an **optimization algorithm**
- Mixing and matching these modular components give us a lot of new ML methods.

Linear Classifiers, Logistic Regression, Multiclass Classification

- **Classification**: predicting a discrete-valued target
 - **Binary classification**: predicting a binary-valued target
 - **Multiclass classification**: predicting a discrete(>2)-valued target
- Examples of binary classification
 - Predict whether a patient has a disease, given the presence or absence of various symptoms
 - Classify e-mails as spam or non-spam
 - Predict whether a financial transaction is fraudulent

Linear Classifiers, Logistic Regression, Multiclass Classification

Binary linear classification

- **Classification:** given a D -dimensional input $\mathbf{x} \in \mathbb{R}^D$ predict a discrete-valued target
- **Binary:** predict a binary target $t \in \{0,1\}$
 - Training examples with $t = 1$ are called **positive examples**, and training examples with $t = 0$ are called **negative examples**. Sorry.
 - $t \in \{0,1\}$ or $t \in \{-1, +1\}$ is for computational convenience.
- **Linear:** model prediction y is a linear function of $\mathbf{x}$, followed by a threshold r :

$$z = \mathbf{w}^\top \mathbf{x} + b$$

$$y = \begin{cases} 1 & \text{if } z \geq r \\ 0 & \text{if } z < r \end{cases}$$

Summary | Binary Linear Classifiers

- **Summary:** targets $t \in \{0,1\}$, inputs $\mathbf{x} \in \mathbb{R}^{D+1}$ with $x_0 = 1$, and model is defined by weights $\mathbf{w}$ and

$$z = \mathbf{w}^\top \mathbf{x}$$

$$y = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

- How can we find good values for $\mathbf{w}$?
- If training set is linearly separable, we could solve for $\mathbf{w}$ using linear programming
 - We could also apply an iterative procedure known as the *perceptron algorithm* (but this is primarily of historical interest).
- If it's not linearly separable, the problem is harder
 - Data is almost never linearly separable in real life.

Logistic Regression

Logistic regression:

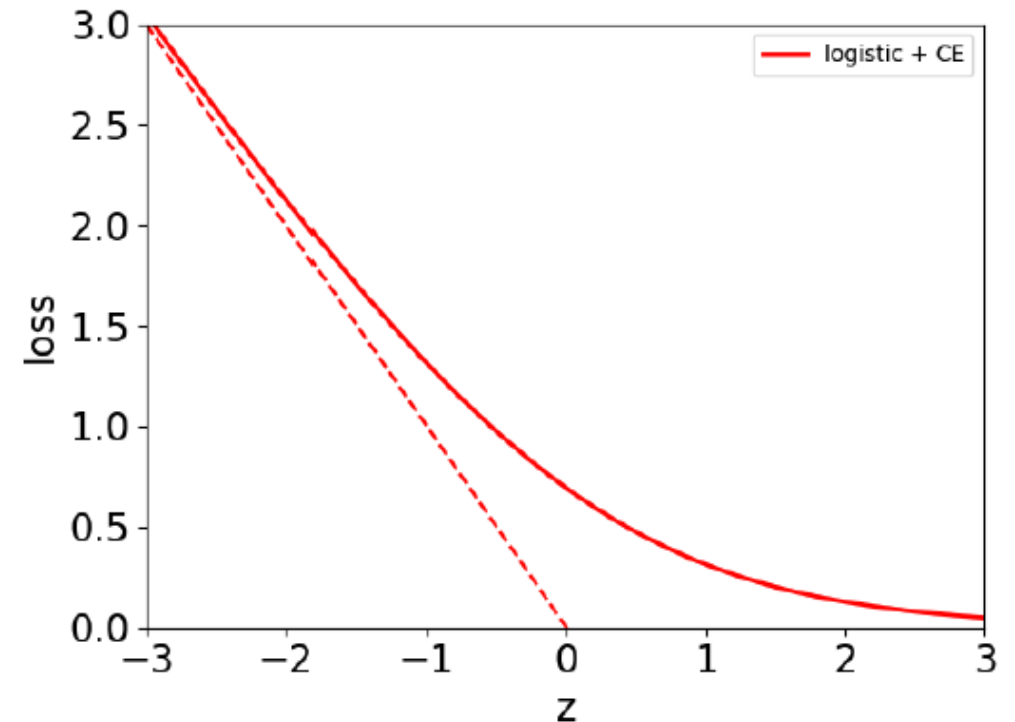
$$z = \mathbf{w}^\top \mathbf{x}$$

$$y = \sigma(z)$$

$$= \frac{1}{1 + e^{-z}}$$

$$\mathcal{L}_{\text{CE}}(y, t) = -t \log y - (1 - t) \log(1 - y)$$

Plot is for target $t = 1$.



Gradient Descent for Logistic Regression

- How do we minimize the cost J for logistic regression? No direct solution.
 - Taking derivatives of J w.r.t. $\mathbf{w}$ and setting them to 0 doesn't have an explicit solution.
- However, the logistic loss is a **convex function** in $\mathbf{w}$, so let's consider the **gradient descent** method from the last lecture.
 - Recall: we **initialize** the weights to something reasonable and repeatedly adjust them in the **direction of steepest descent**.
 - A standard initialization is $\mathbf{w} = 0$. (why?)

Gradient of Logistic Loss

Back to logistic regression:

$$\mathcal{L}_{\text{CE}}(y, t) = -t \log y - (1 - t) \log(1 - y)$$

$$y = 1/(1 + e^{-z}) \text{ and } z = \mathbf{w}^T \mathbf{x}$$

Therefore

$$\begin{aligned} \frac{\partial \mathcal{L}_{\text{CE}}}{\partial w_j} &= \frac{\partial \mathcal{L}_{\text{CE}}}{\partial y} \cdot \frac{\partial y}{\partial z} \cdot \frac{\partial z}{\partial w_j} = \left(-\frac{t}{y} + \frac{1-t}{1-y} \right) \cdot y(1-y) \cdot x_j \\ &= (y - t)x_j \end{aligned}$$

(verify this)

Gradient descent (coordinatewise) update to find the weights of logistic regression:

$$\begin{aligned} w_j &\leftarrow w_j - \alpha \frac{\partial \mathcal{J}}{\partial w_j} \\ &= w_j - \frac{\alpha}{N} \sum_{i=1}^N (y^{(i)} - t^{(i)}) x_j^{(i)} \end{aligned}$$

Gradient Descent for Logistic Regression

Comparison of gradient descent updates:

- Linear regression:

$$\mathbf{w} \leftarrow \mathbf{w} - \frac{\alpha}{N} \sum_{i=1}^N (y^{(i)} - t^{(i)}) \mathbf{x}^{(i)}$$

- Logistic regression:

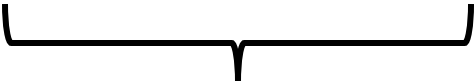
$$\mathbf{w} \leftarrow \mathbf{w} - \frac{\alpha}{N} \sum_{i=1}^N (y^{(i)} - t^{(i)}) \mathbf{x}^{(i)}$$

- Not a coincidence! These are both examples of **generalized linear models**. But we won't go in further detail.
- Notice $\frac{1}{N}$ in front of sums due to averaged losses. This is why you need smaller learning rate when cost is summed losses ($\alpha' = \alpha/N$).

Multiclass Classification

- Targets form a discrete set $\{1, \dots, K\}$.
- It's often more convenient to represent them as **one-hot vectors**, or a **one-of-K encoding**:

$$\mathbf{t} = (0, \dots, 0, 1, 0 \dots, 0) \in \mathbb{R}^K$$


entry k is 1

Multiclass Linear Classification

- We can start with a linear function of the inputs.
- Now there are D input dimensions and K output dimensions, so we need $K \times D$ weights, which we arrange as a **weight matrix $\mathbf{W}$** .
- Also, we have a K -dimensional vector **$\mathbf{b}$** of biases.
- A linear function of the inputs:

$$z_k = \sum_{j=1}^D w_{kj} x_j + b_k \quad \text{for } k = 1, 2, \dots, K$$

- We can eliminate the bias **$\mathbf{b}$** by taking **$\mathbf{W} \in \mathbb{R}^{K \times (D+1)}$** and adding a dummy variable $x_0 = 1$. So, vectorized:

$$\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b} \quad \text{or with dummy } x_0 = 1 \quad \mathbf{z} = \mathbf{W}\mathbf{x}$$

Multiclass Linear Classification

- How can we turn this linear prediction into a **one-hot prediction**?
- We can interpret the magnitude of z_k as a measure of how much the model prefers k as its prediction.
- If we do this, we should set

$$y_i = \begin{cases} 1 & i = \arg \max_k z_k \\ 0 & \text{otherwise} \end{cases}$$

Softmax Regression

- We need to soften our predictions for the sake of optimization.
- We want soft predictions that are like probabilities, i.e., $0 \leq y_k \leq 1$ and $\sum_k y_k = 1$.
- A natural activation function to use is the **softmax function**, a multivariable generalization of the logistic function:

$$y_k = \text{softmax}(z_1, \dots, z_K)_k = \frac{e^{z_k}}{\sum_{k'} e^{z_{k'}}}$$

- Outputs can be interpreted as probabilities (positive and sum to 1)
- If z_k is much larger than the others, then $\text{softmax}(\mathbf{z})_k \approx 1$ and it behaves like argmax .

Softmax Regression

- If a model outputs a vector of class probabilities, we can use cross-entropy as the loss function:

$$\begin{aligned}\mathcal{L}_{\text{CE}}(\mathbf{y}, \mathbf{t}) &= - \sum_{k=1}^K t_k \log y_k \\ &= -\mathbf{t}^\top (\log \mathbf{y}),\end{aligned}$$

where the log is applied elementwise.

- Just like with logistic regression, we typically combine the softmax and cross-entropy into a **softmax-cross-entropy** function.

Softmax Regression

- Softmax regression (with dummy $x_0 = 1$):

$$\mathbf{z} = \mathbf{W}\mathbf{x}$$

$$\mathbf{y} = \text{softmax}(\mathbf{z})$$

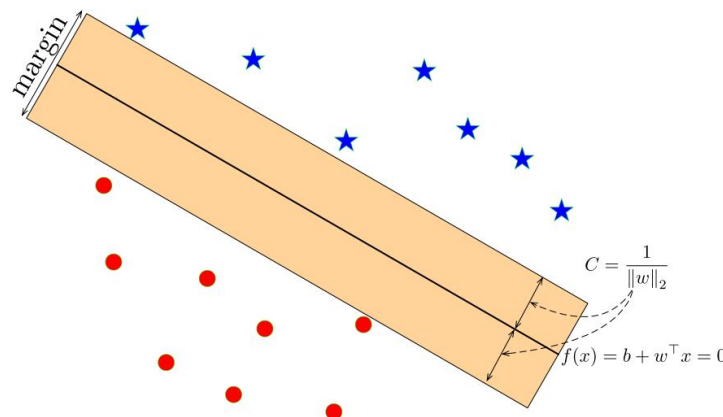
$$\mathcal{L}_{\text{CE}} = -\mathbf{t}^T(\log \mathbf{y})$$

- Gradient descent updates can be derived for each row of $\mathbf{W}$:

$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial \mathbf{w}_k} = \frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_k} \cdot \frac{\partial z_k}{\partial \mathbf{w}_k} = (y_k - t_k) \cdot \mathbf{x}$$

$$\mathbf{w}_k \leftarrow \mathbf{w}_k - \alpha \frac{1}{N} \sum_{i=1}^N \left(y_k^{(i)} - t_k^{(i)} \right) \mathbf{x}^{(i)}$$

- Similar to linear/logistic reg (no coincidence) (verify the update)



A **Support Vector Machine (SVM)** is a supervised machine learning algorithm used for classification and regression tasks. It is particularly effective in high-dimensional spaces and is well-suited for scenarios where there is a clear margin of separation between classes.

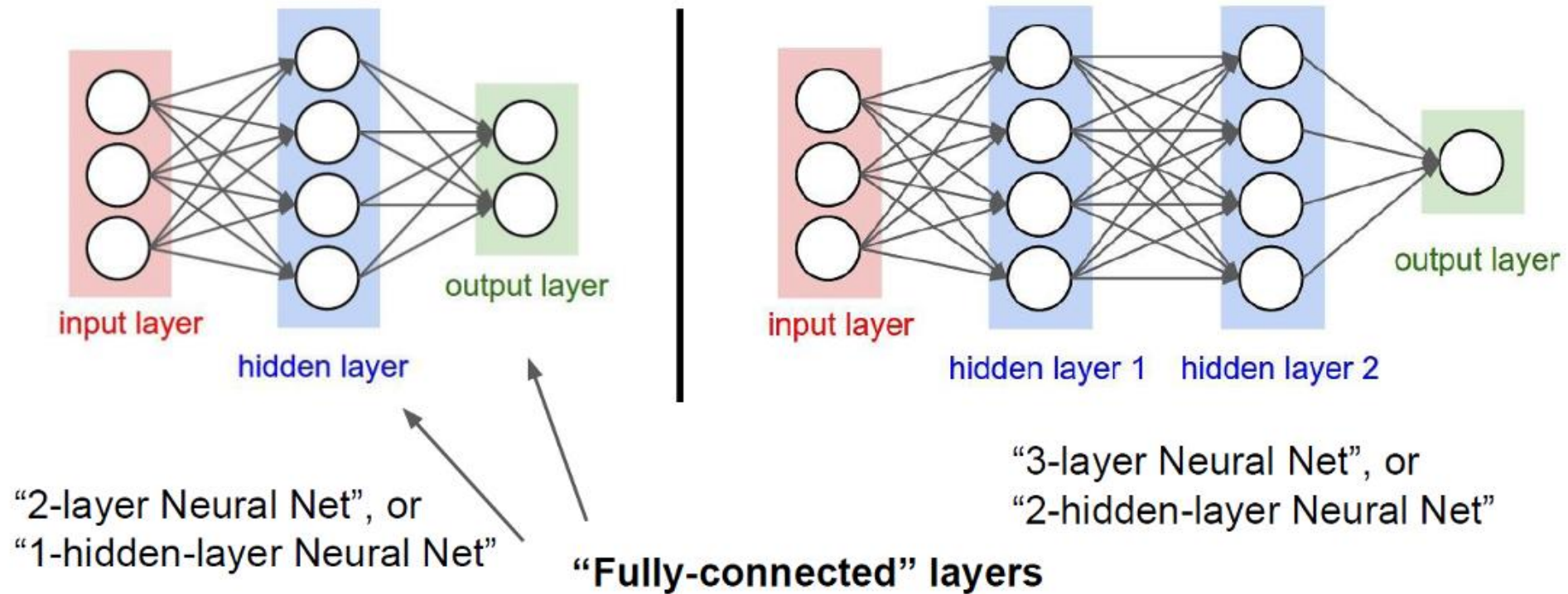
The basic idea behind SVM is to find the hyperplane that **best divides a dataset into two classes**.

The term "**support vector**" refers to the data points that lie closest to the decision boundary, and these points are crucial in determining the optimal hyperplane.

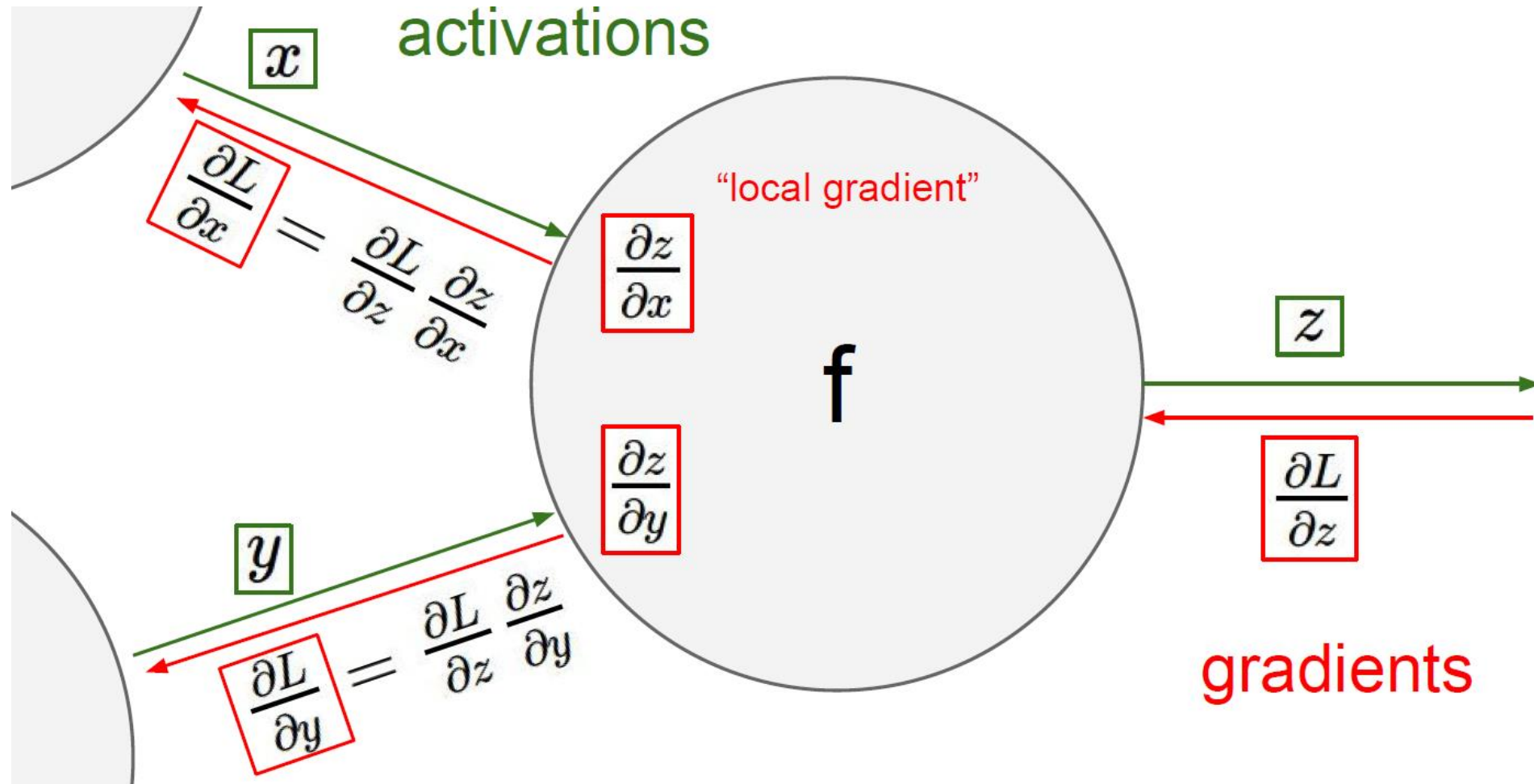
The **goal** is to maximize the margin, which is the distance between the hyperplane and the nearest data points of each class.

Neural network

Neural Networks: Architectures



Chain rule



Gradients for vector

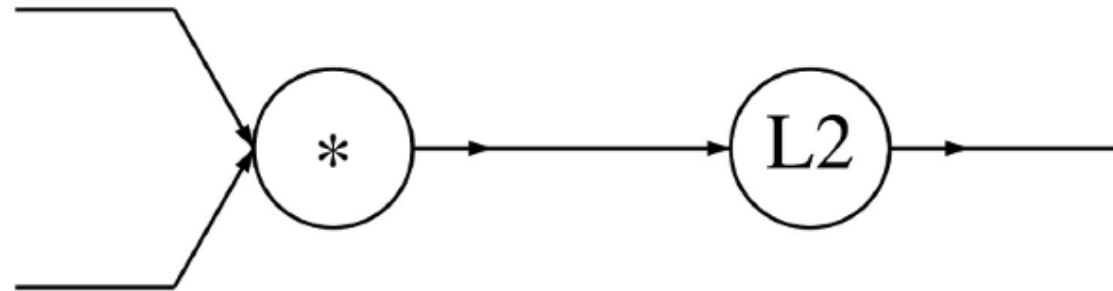
A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

$$\begin{bmatrix} 0.1 & 0.5 \\ -0.3 & 0.8 \end{bmatrix}$$

W

$$\begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

x

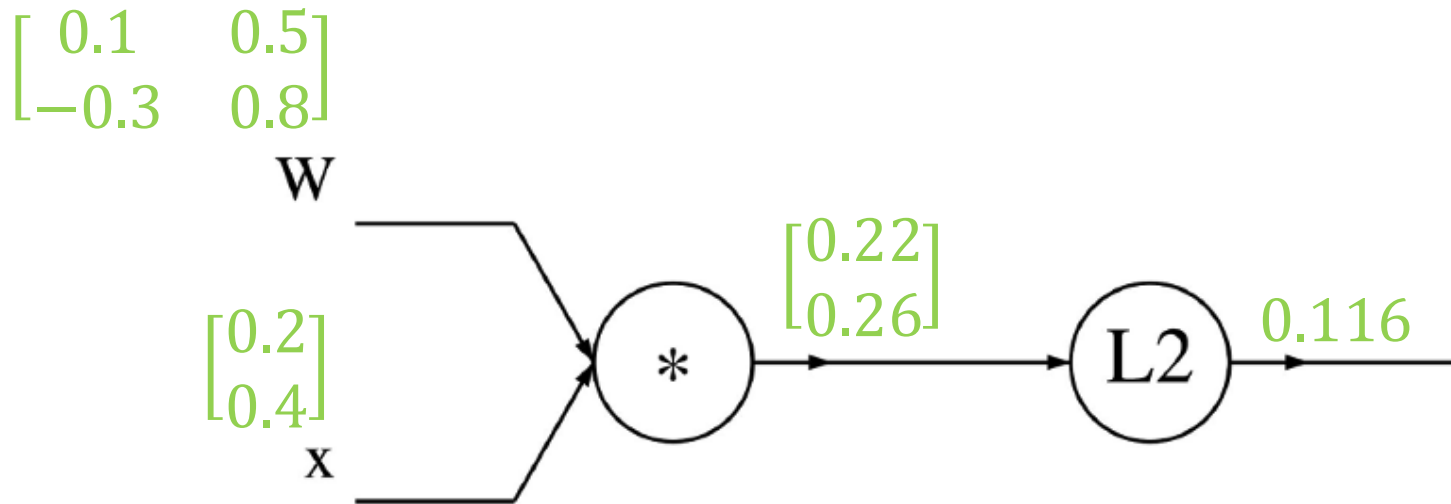


$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \cdots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \cdots + W_{n,n}x_n \end{pmatrix}$$

$$f(q) = \|q\|^2 = q_1^2 + \cdots + q_n^2$$

Gradients for vector

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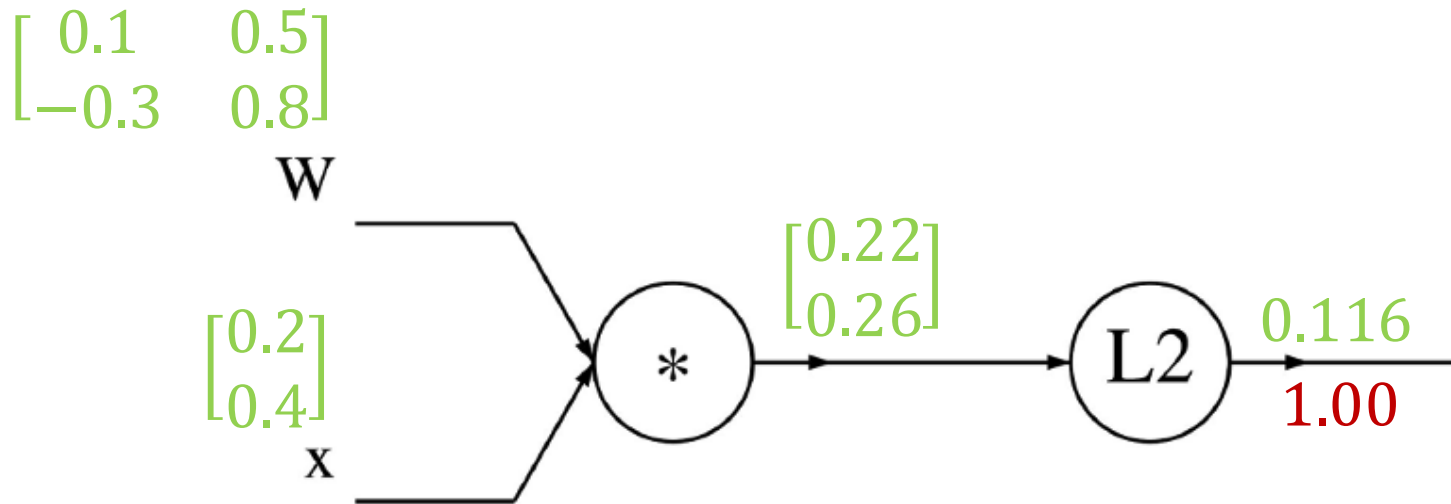


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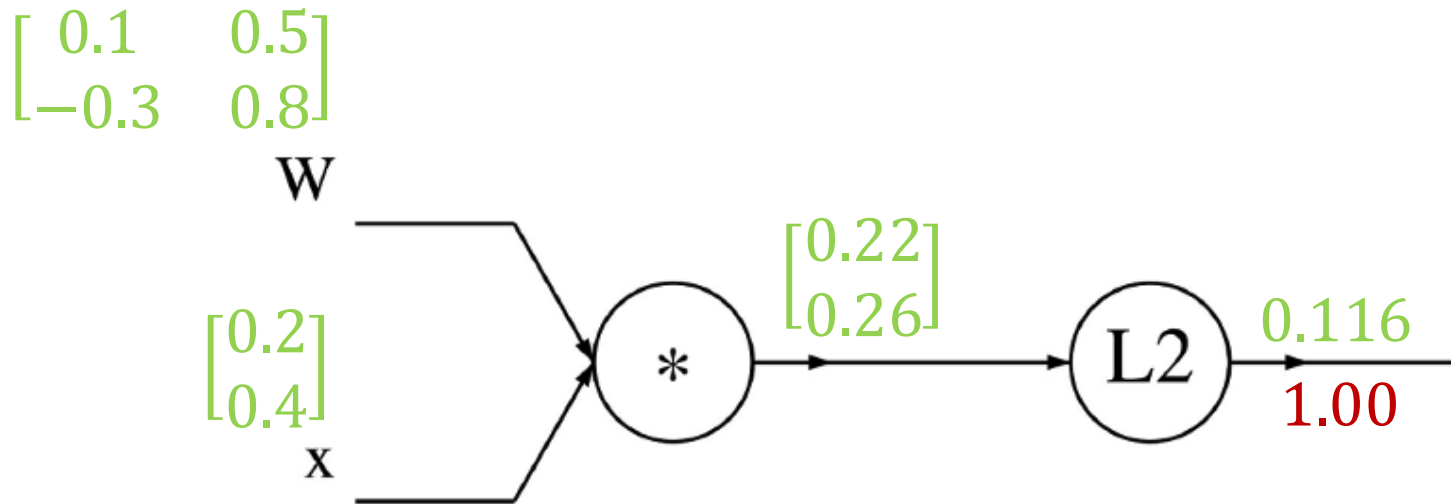


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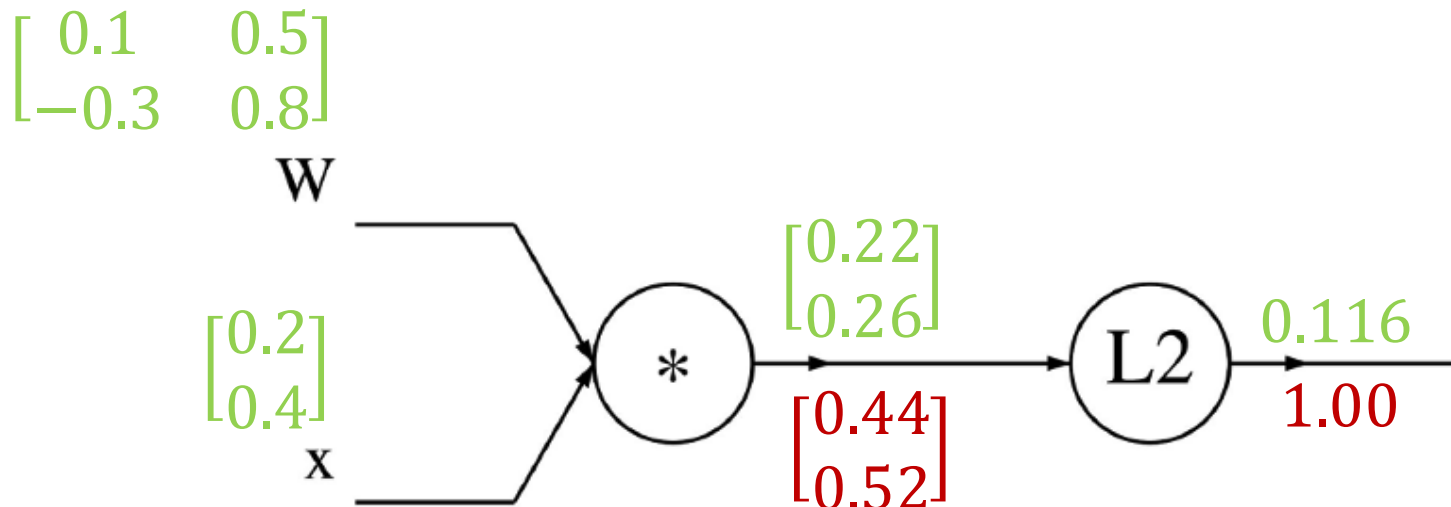
$$\frac{\partial f}{\partial q_i} = 2q_i$$

$$\boxed{\nabla_q f = 2q}$$

$$f(q) = \|q\|^2 = q_1^2 + \dots + q_n^2$$

Gradients for vector

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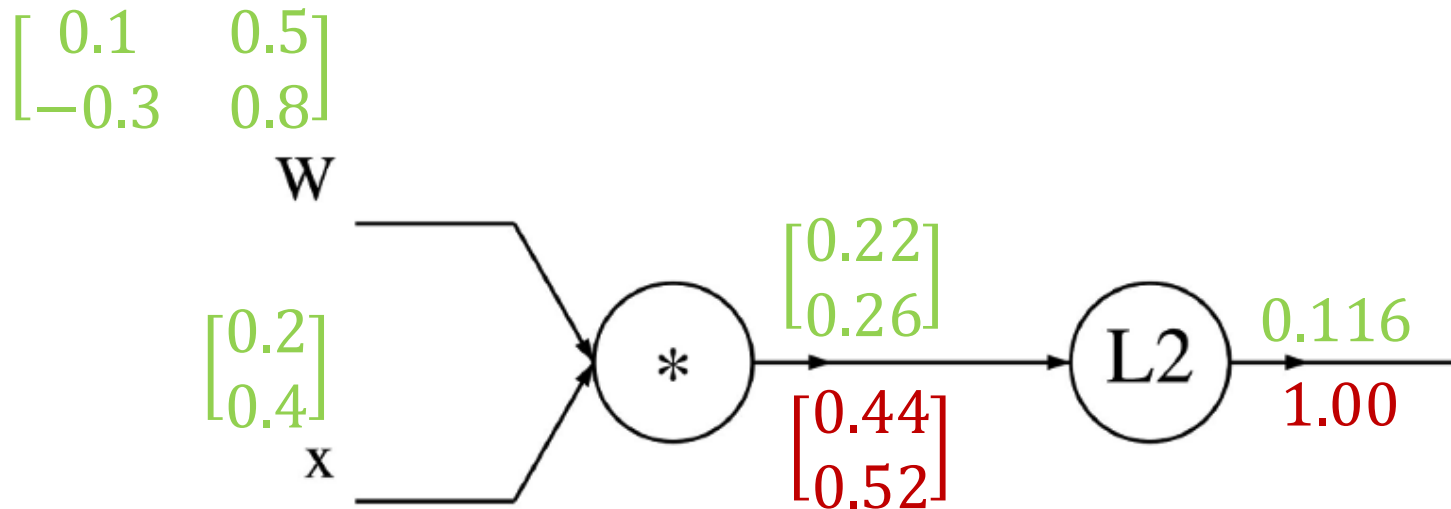
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$$\frac{\partial q_k}{\partial W_{i,j}} = \mathbf{1}_{k=i} x_j$$

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Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

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W

$$\begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$$

x

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$$\begin{bmatrix} 0.44 \\ 0.52 \end{bmatrix}$$

L2

$$0.116$$

$$1.00$$

$$q = W \cdot x = \begin{pmatrix} W_{1,1}x_1 + \cdots + W_{1,n}x_n \\ \vdots \\ W_{n,1}x_1 + \cdots + W_{n,n}x_n \end{pmatrix}$$

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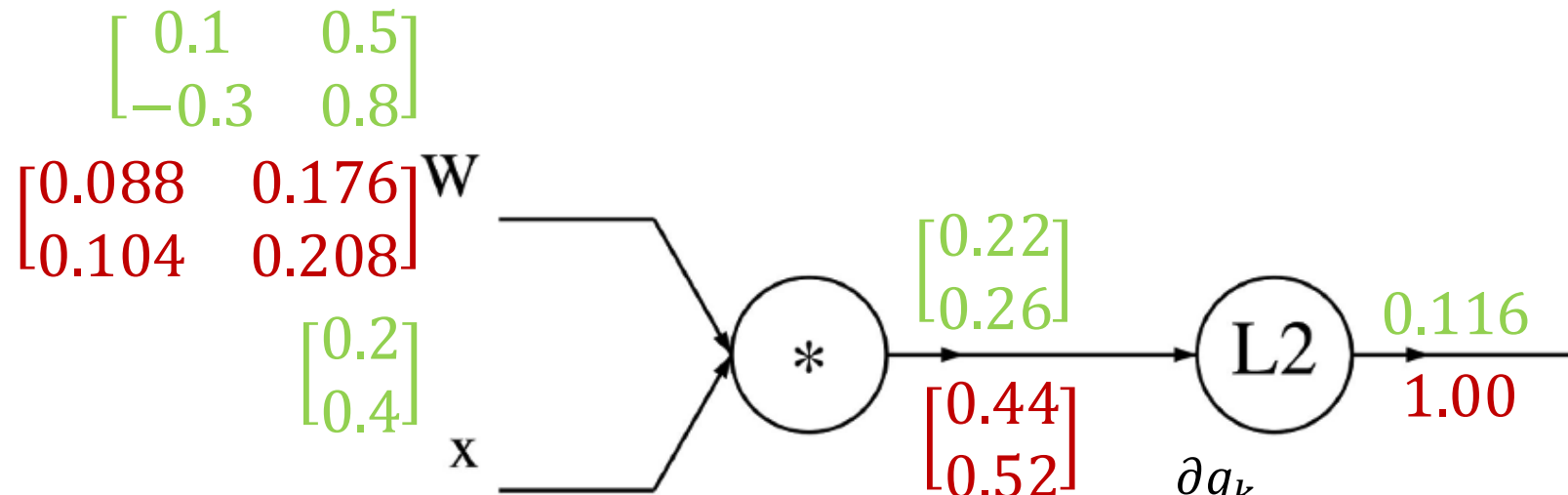
$$\frac{\partial q_k}{\partial W_{i,j}} = \mathbf{1}_{k=i} x_j$$

$$\frac{\partial f}{\partial W_{i,j}} = \sum_k \frac{\partial f}{\partial q_k} \frac{\partial q_k}{\partial W_{i,j}}$$

$$= \sum_k (2q_k)(\mathbf{1}_{k=i} x_j)$$

Gradients for vector

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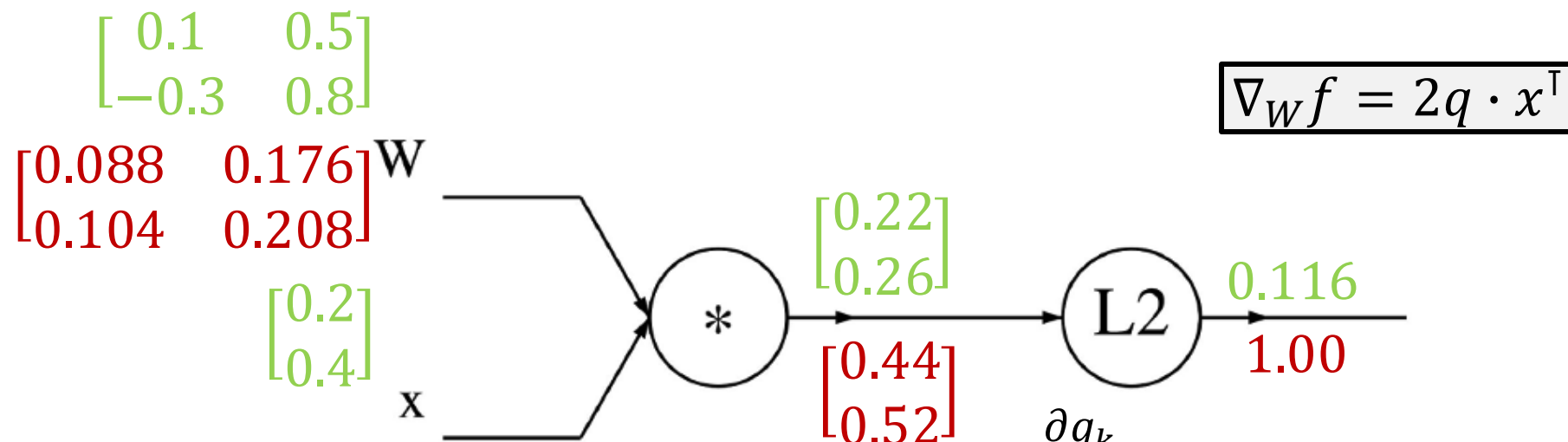
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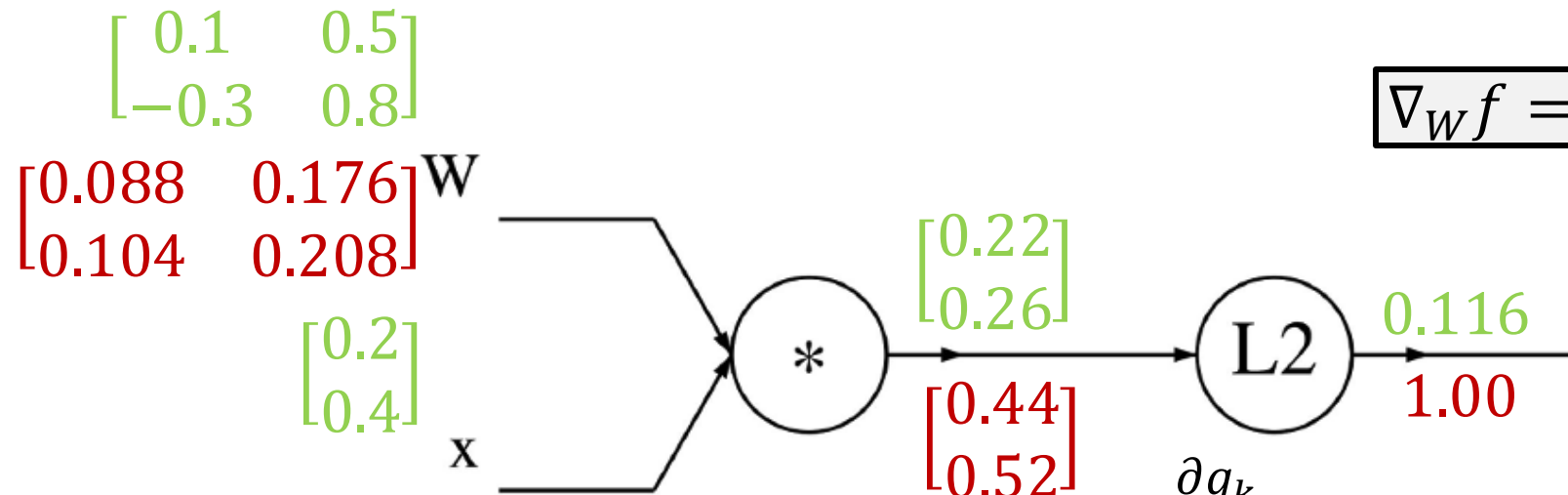
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Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$



$$\nabla_W f = 2q \cdot x^T$$

Always check: The gradient with respect to a variable should have the same shape as the variable

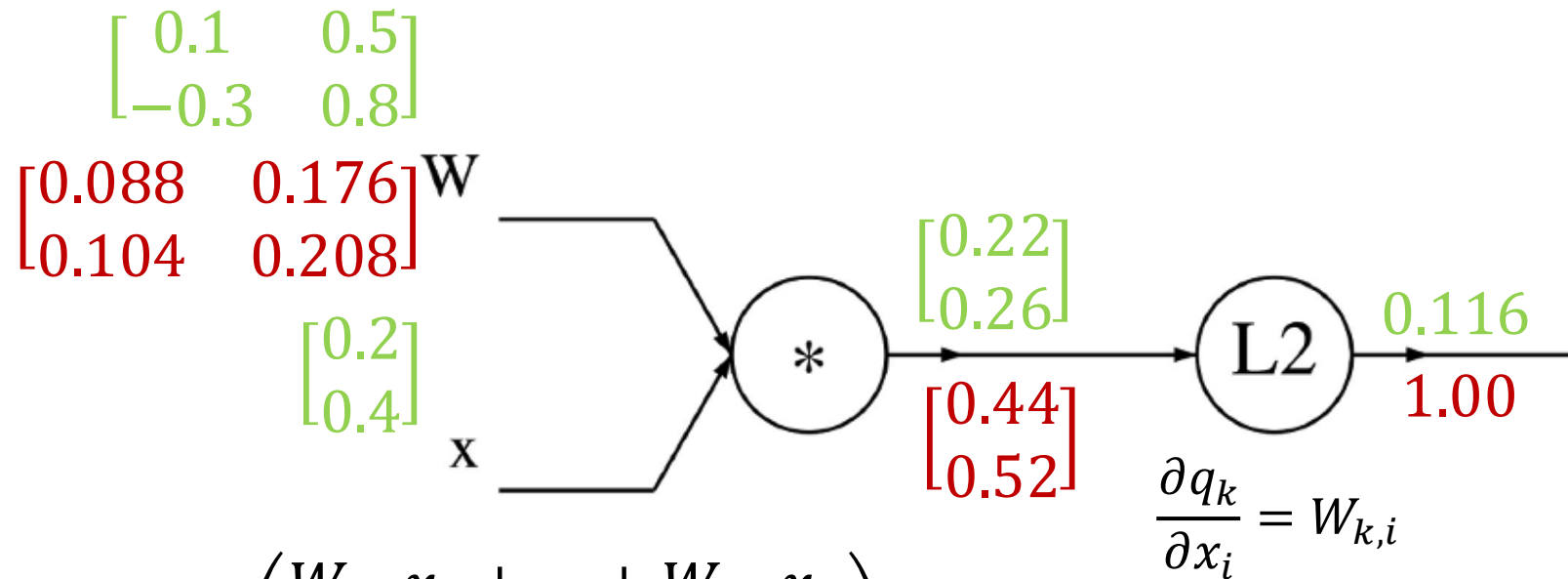
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Gradients for vector

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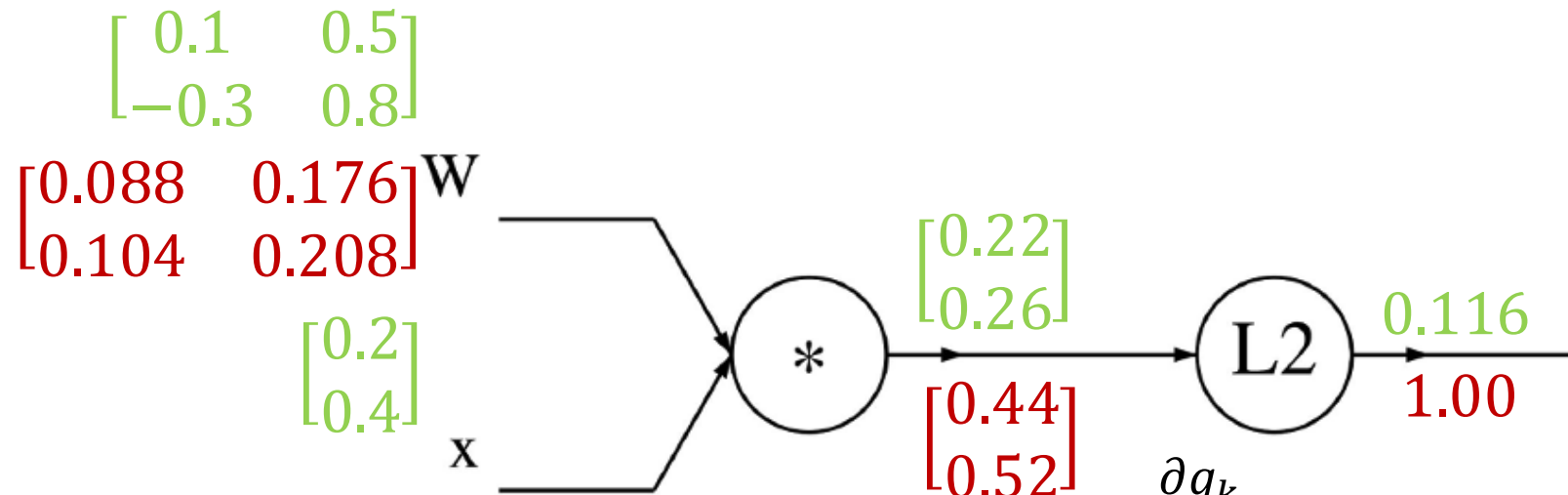


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Gradients for vector

A vectorized example: $f(x, W) = \|W \cdot x\|^2 = \sum_{i=1}^n (W \cdot x)_i^2$

$$\begin{bmatrix} 0.1 & 0.5 \\ -0.3 & 0.8 \end{bmatrix}$$

$$\begin{bmatrix} 0.088 & 0.176 \\ 0.104 & 0.208 \end{bmatrix}^W$$

$$\begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}^x$$

$$\begin{bmatrix} -0.112 \\ 0.636 \end{bmatrix}$$

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$$f(q) = \|q\|^2 = q^2 + \dots + q^2$$

$$\boxed{\nabla_x f = 2W^T \cdot q}$$

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$$\text{L2}$$

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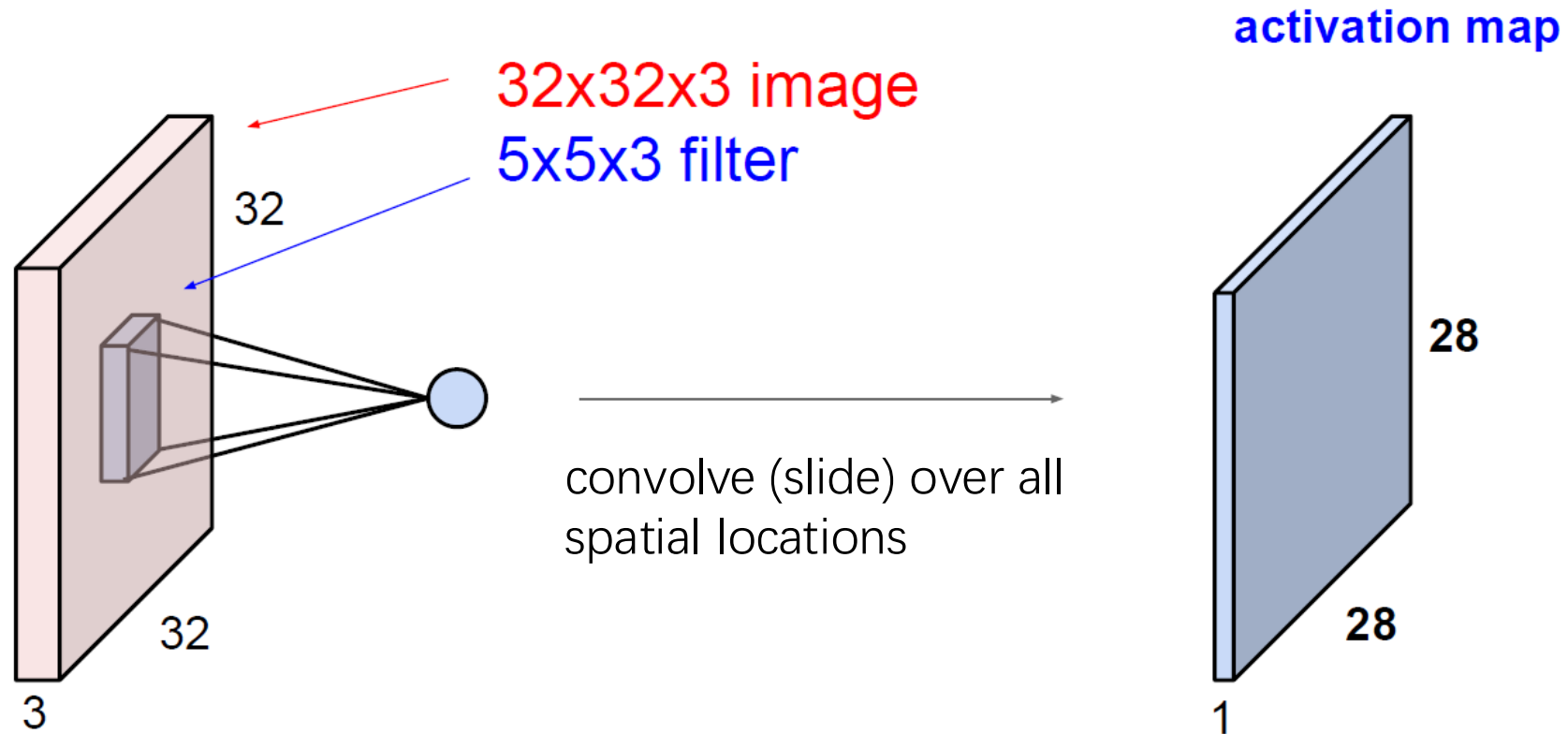
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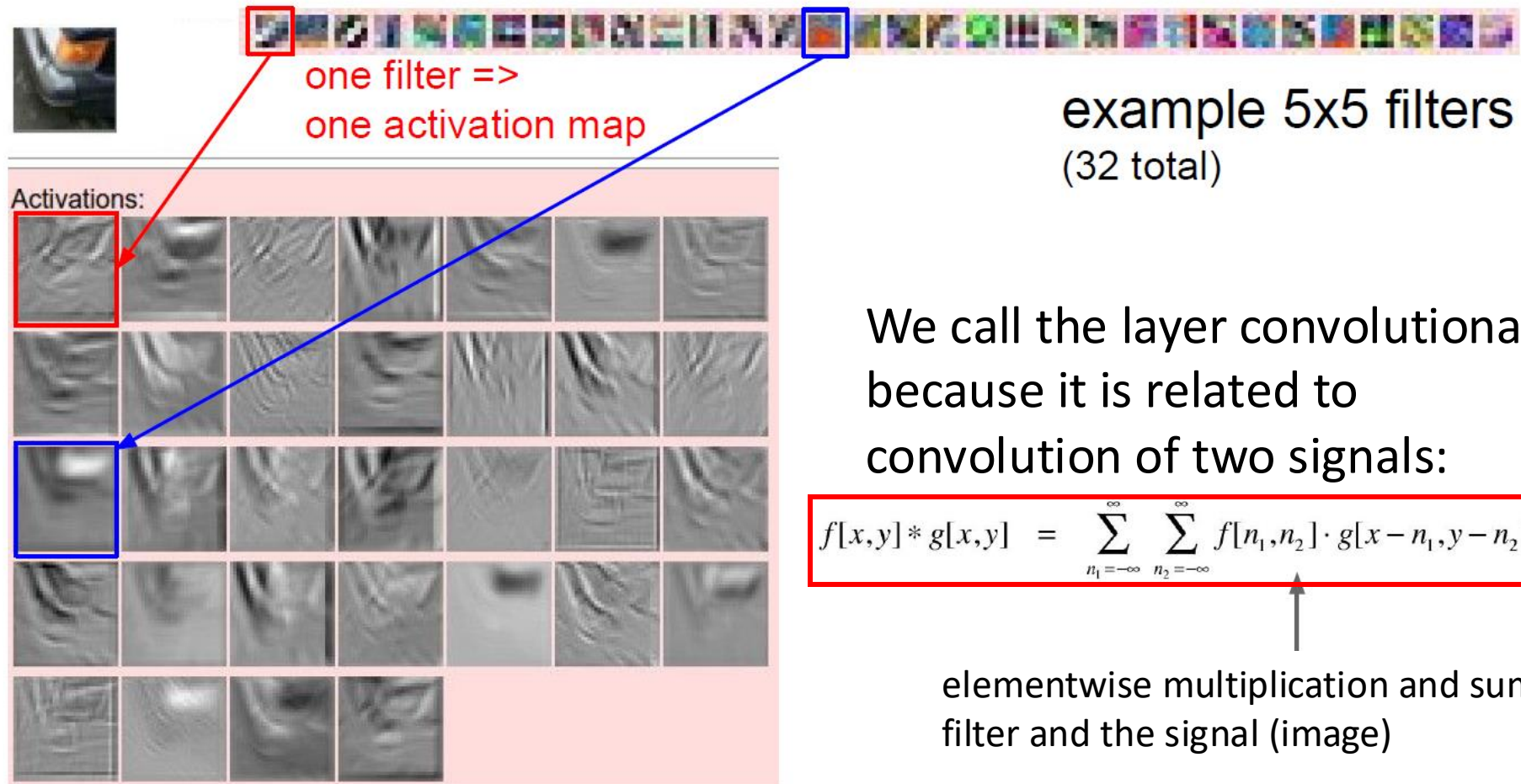
$$= \sum_k 2q_k W_{k,i}$$

CNN

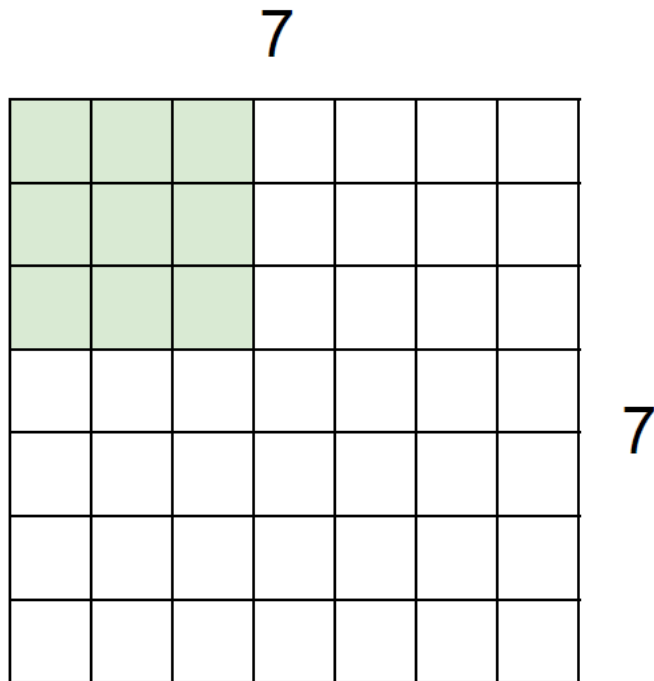
A closer look at spatial dimensions:



CNN

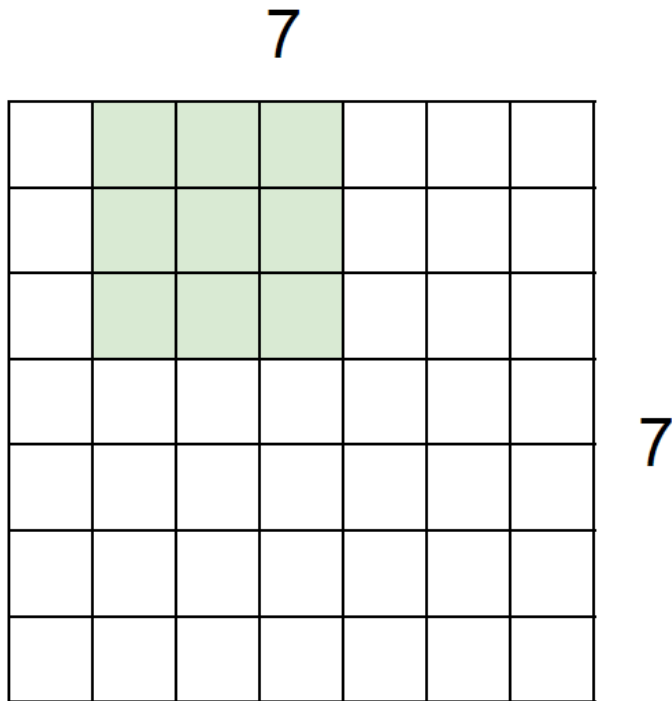


A closer look at spatial dimensions:



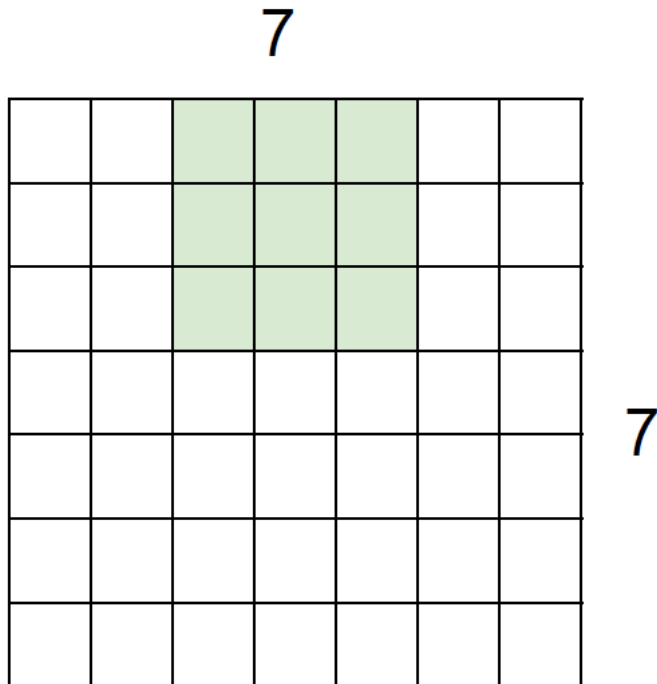
7x7 input (spatially)
assume 3x3 filter

A closer look at spatial dimensions:



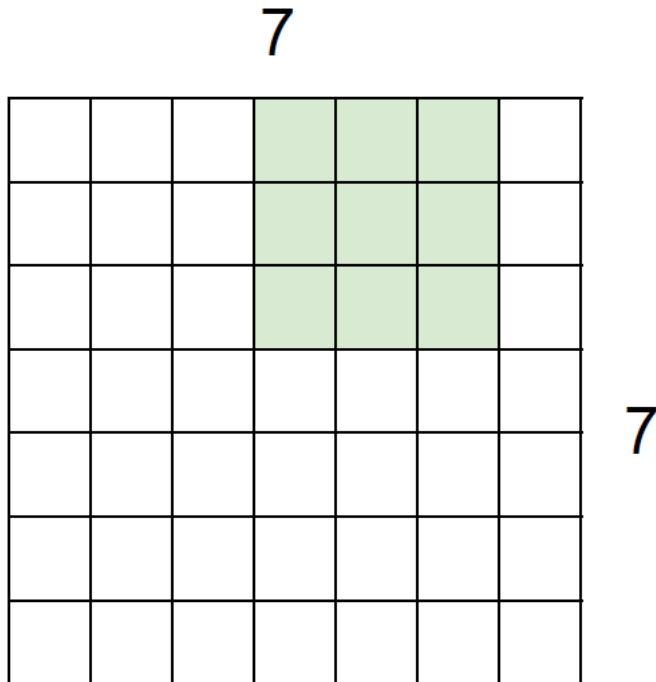
7x7 input (spatially)
assume 3x3 filter

A closer look at spatial dimensions:



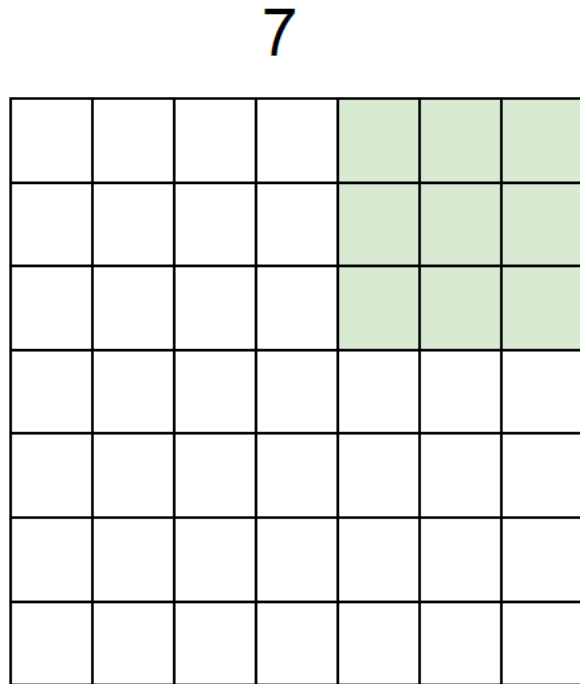
7x7 input (spatially)
assume 3x3 filter

A closer look at spatial dimensions:



7x7 input (spatially)
assume 3x3 filter

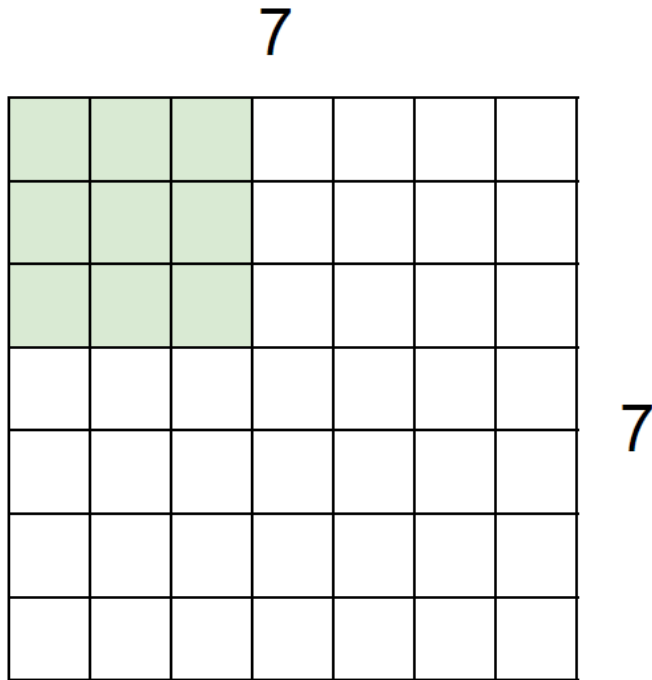
A closer look at spatial dimensions:



7x7 input (spatially)
assume 3x3 filter

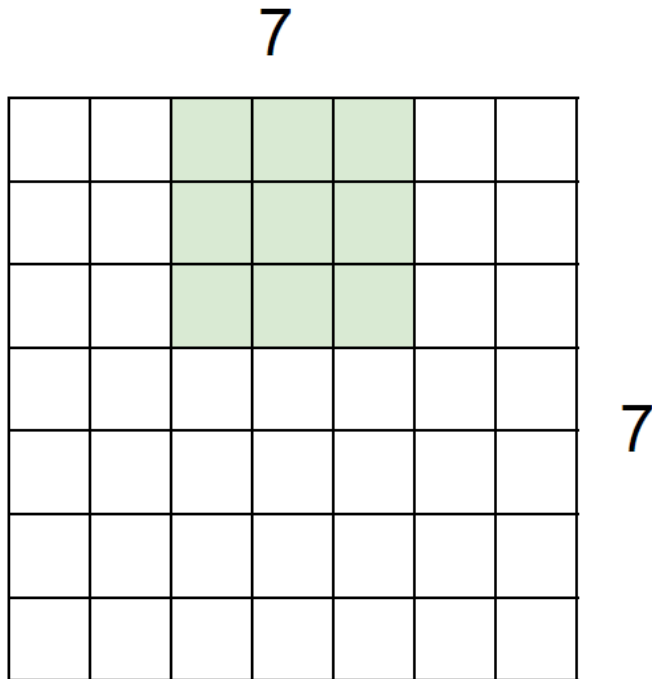
=> **5x5 output**

A closer look at spatial dimensions:



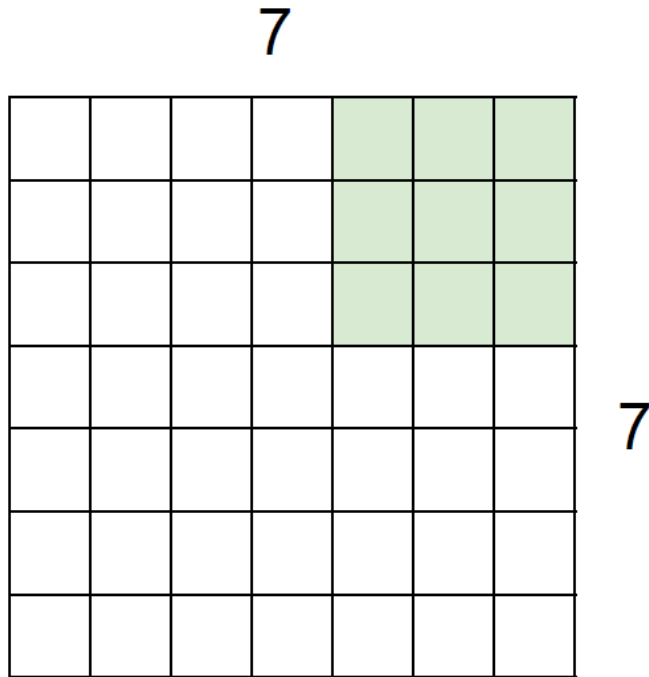
7x7 input (spatially)
assume 3x3 filter
applied **with stride 2**

A closer look at spatial dimensions:



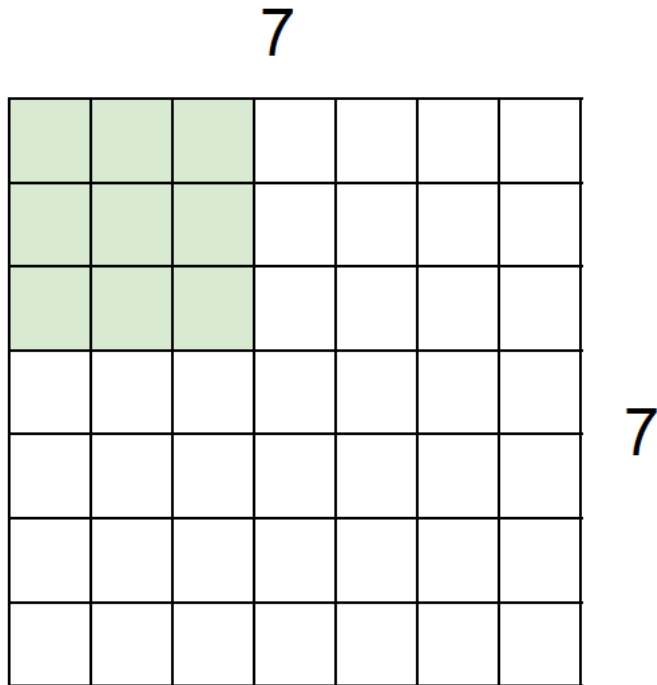
7x7 input (spatially)
assume 3x3 filter
applied **with stride 2**

A closer look at spatial dimensions:



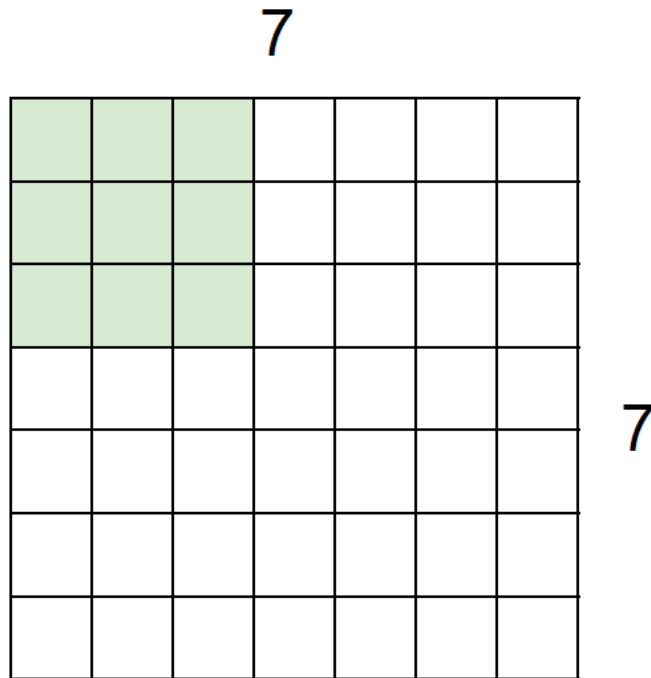
7x7 input (spatially)
assume 3x3 filter
applied **with stride 2**
=> 3x3 output!

A closer look at spatial dimensions:



7x7 input (spatially)
assume 3x3 filter
applied **with stride 3?**

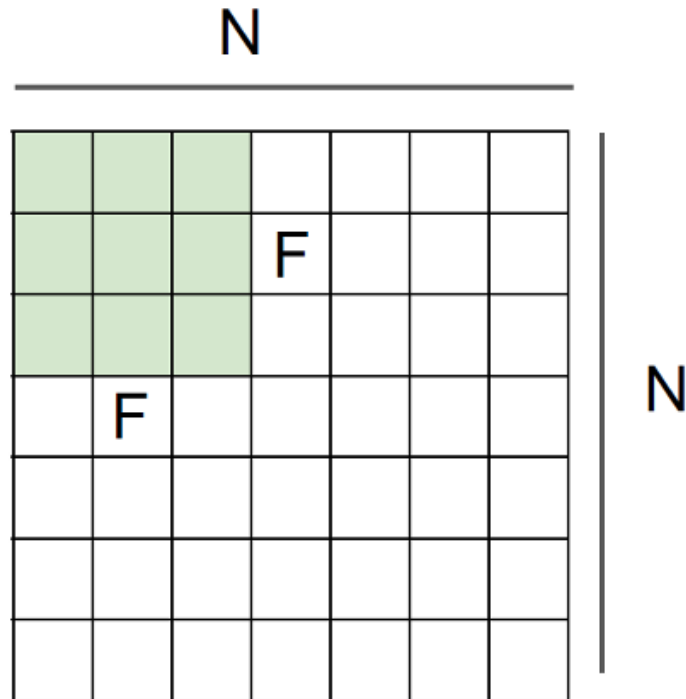
A closer look at spatial dimensions:



7x7 input (spatially)
assume 3x3 filter
applied **with stride 3?**

doesn't fit!
Cannot apply 3x3 filter on 7x7
input with stride 3.

CNN



Output size:
 $(N - F) / \text{stride} + 1$

e.g., $N=7, F=3$:

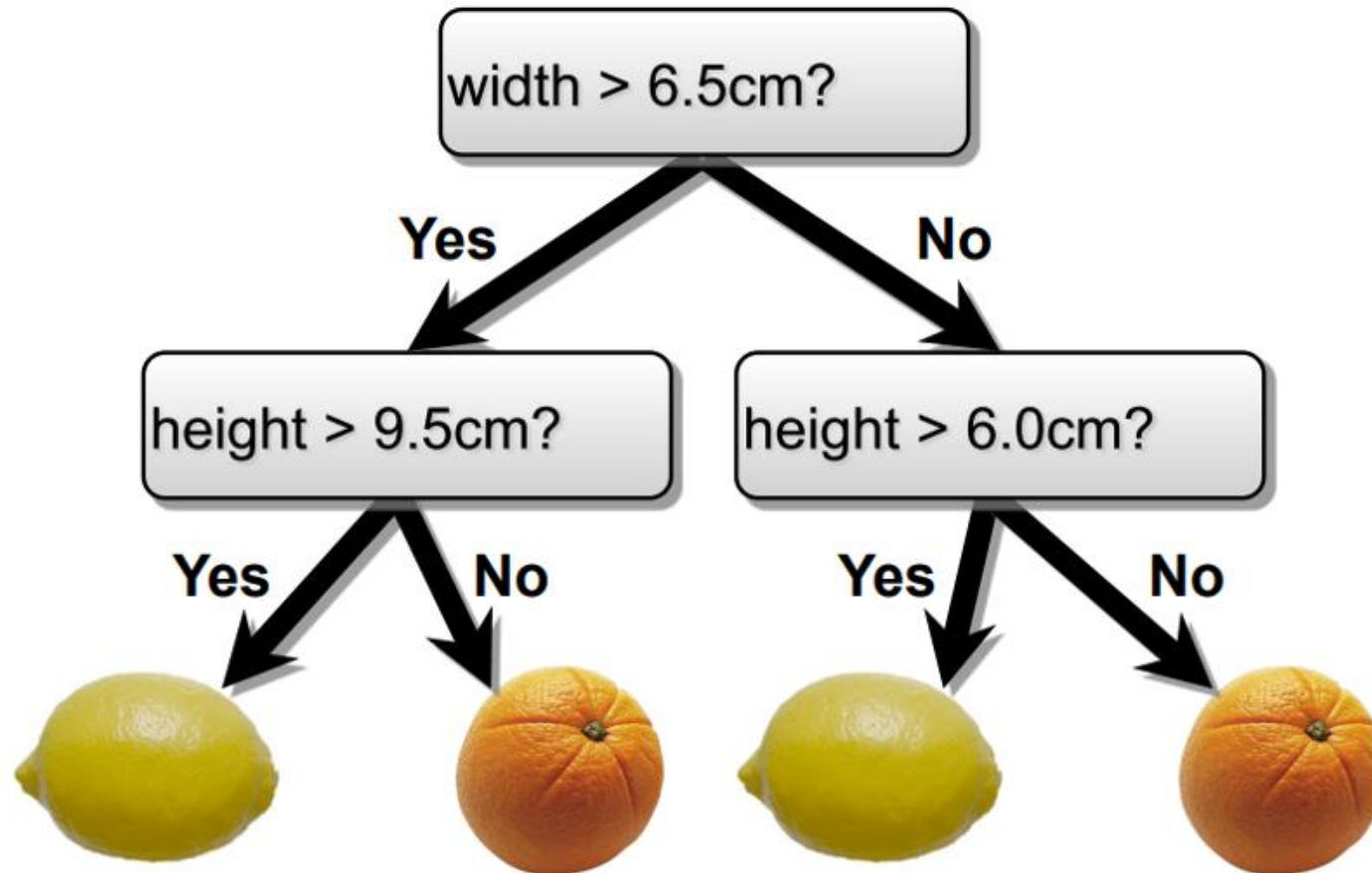
stride 1 $\Rightarrow (7 - 3) / 1 + 1 = 5$

stride 2 $\Rightarrow (7 - 3) / 2 + 1 = 3$

stride 3 $\Rightarrow (7 - 3) / 3 + 1 = 2.33 : \backslash$

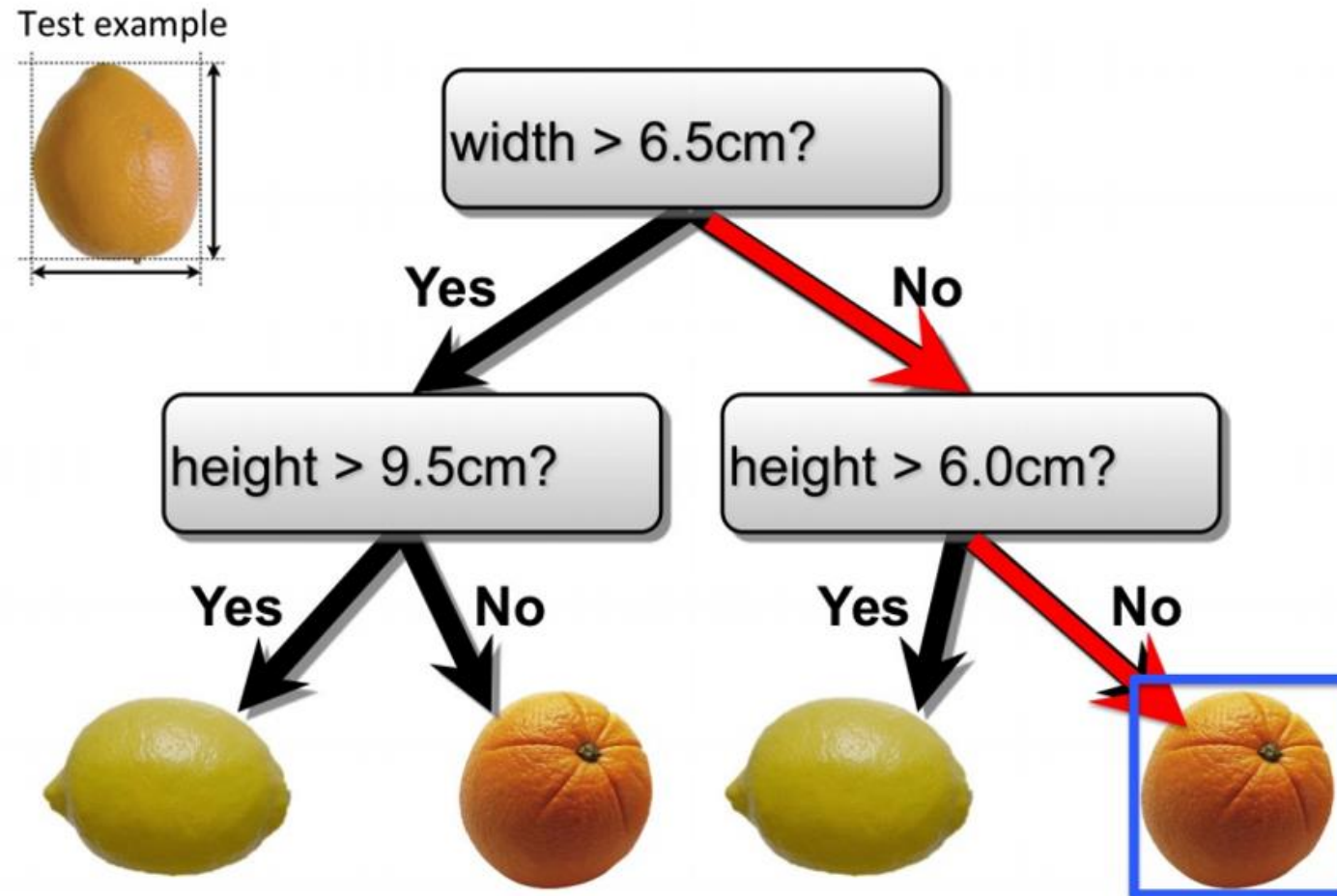
Decision Trees

- Make predictions by splitting on features according to a tree structure.



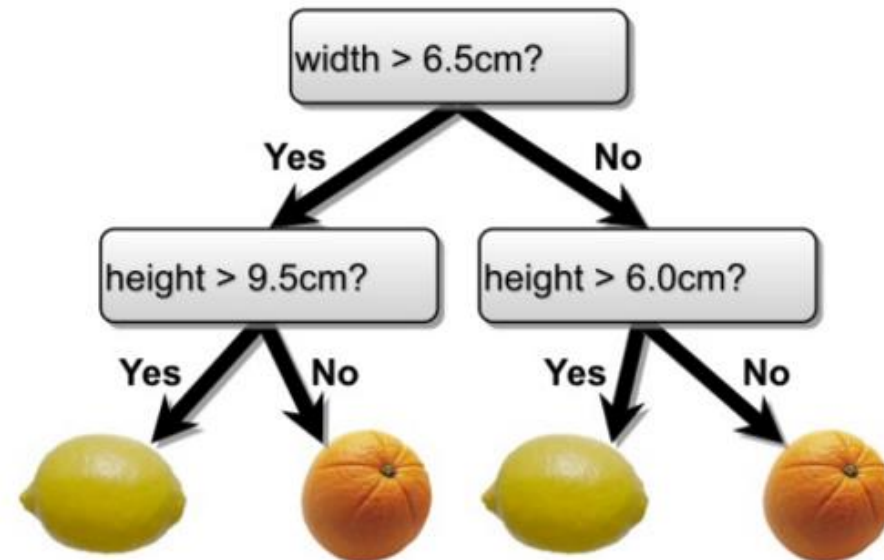
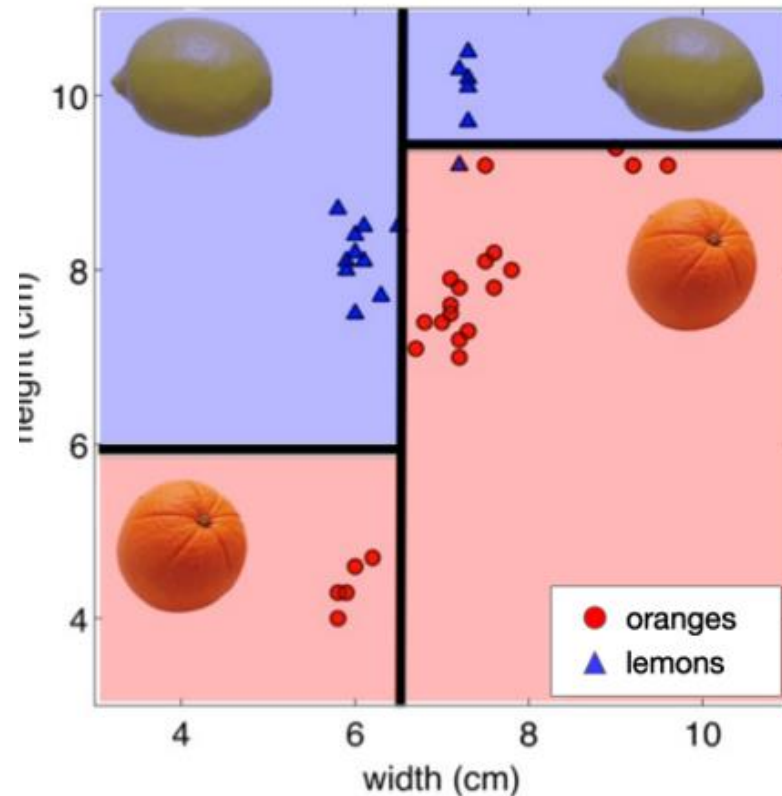
Decision Trees

- Make predictions by splitting on features according to a tree structure.

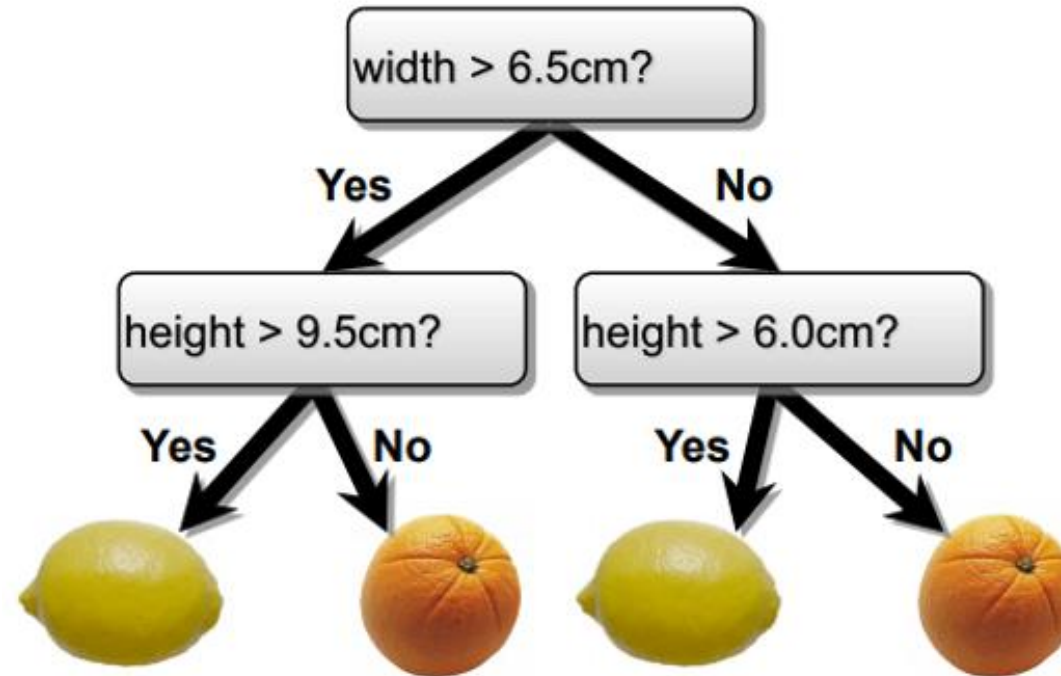


Decision Trees— Continuous Features

- Split *continuous features* by checking whether that feature is greater than or less than some threshold.
- Decision boundary is made up of axis-aligned planes.



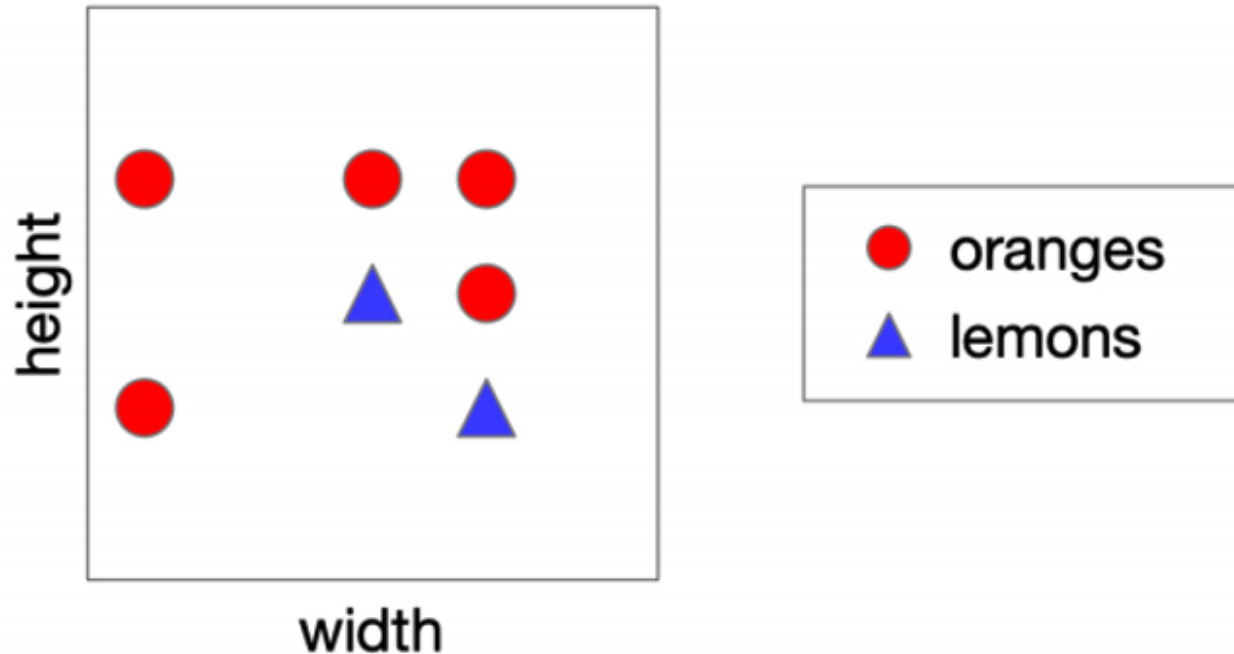
Decision Trees



- Internal nodes test a feature
- Branching is determined by the feature value
- Leaf nodes are outputs (predictions)

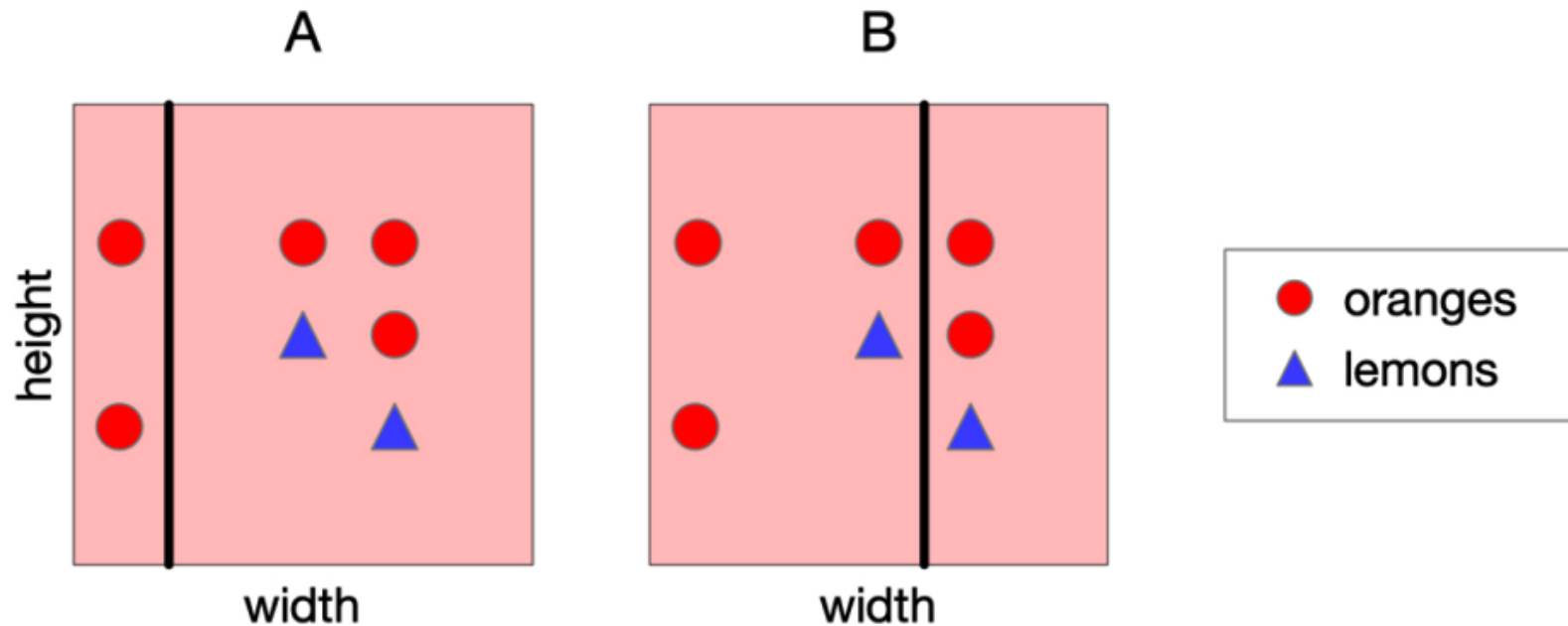
Decision Tree-Examples

- Consider the following data. Let's split on width.



Choosing a Good Split

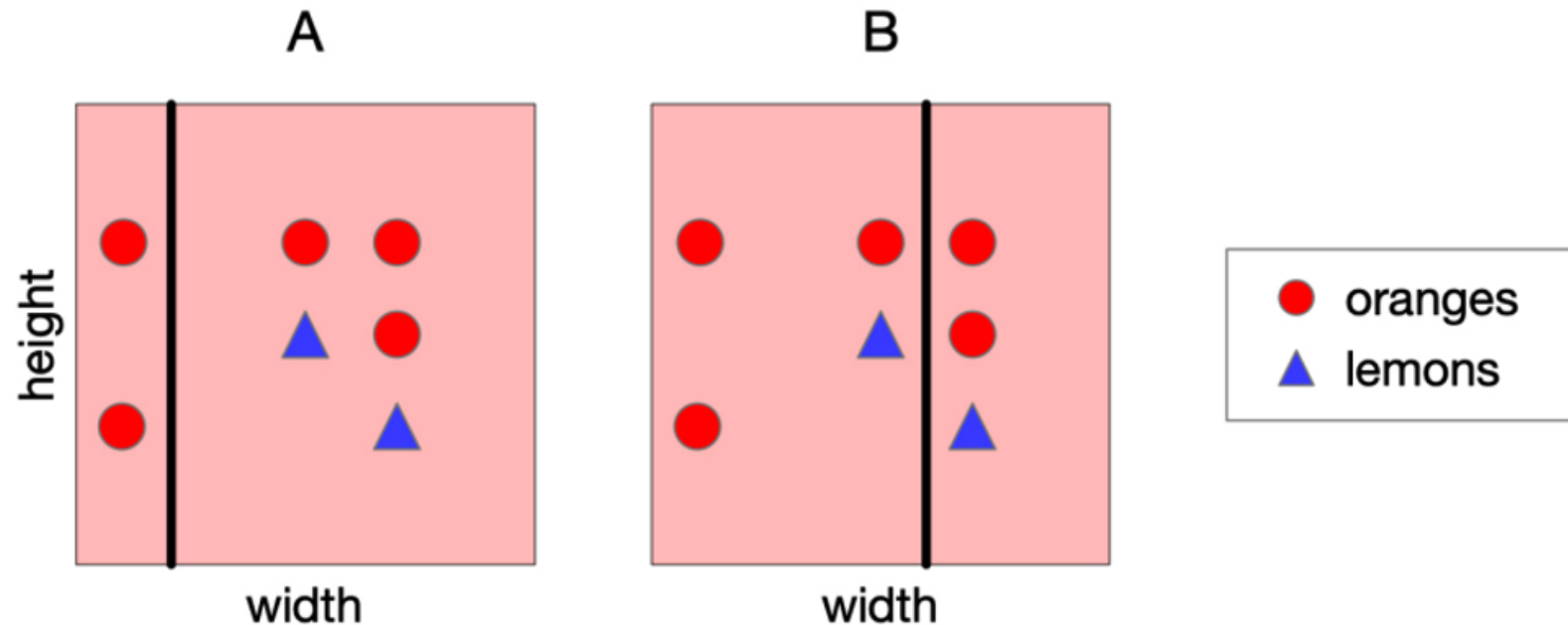
- Recall: classify by majority.



- A and B have the same misclassification rate, so which is the best split? Vote!

Choosing a Good Split

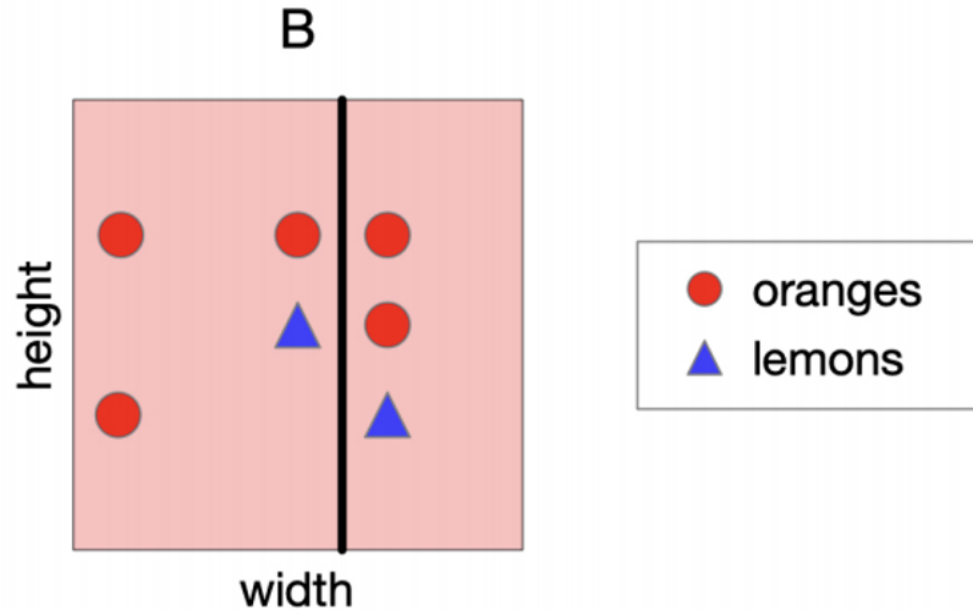
- A feels like a better split, because the left-hand region is very certain about whether the fruit is an orange.



- Can we quantify this?

Revisiting Our Original Example

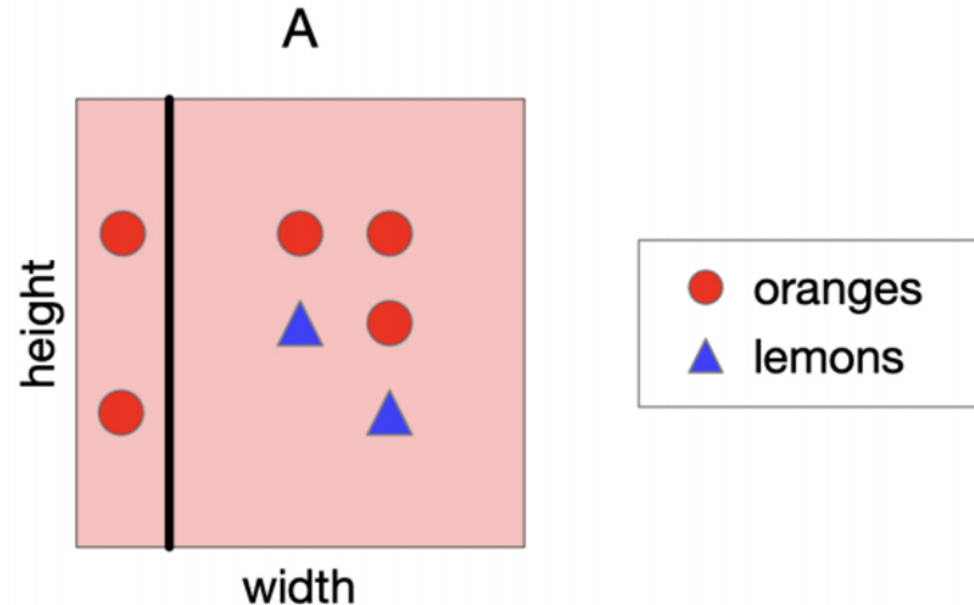
- What is the information gain of split B? Not terribly informative...



- Root entropy of class outcome: $H(Y) = -\frac{2}{7}\log_2\left(\frac{2}{7}\right) - \frac{5}{7}\log_2\left(\frac{5}{7}\right) \approx 0.86$
- Leaf conditional entropy of class outcome: $H(Y|left) \approx 0.81$, $H(Y|right) \approx 0.92$
- $IG(split) \approx 0.86 - \left(\frac{4}{7} \cdot 0.81 + \frac{3}{7} \cdot 0.92\right) \approx 0.006$

Revisiting Our Original Example

- What is the information gain of split A? Very informative!

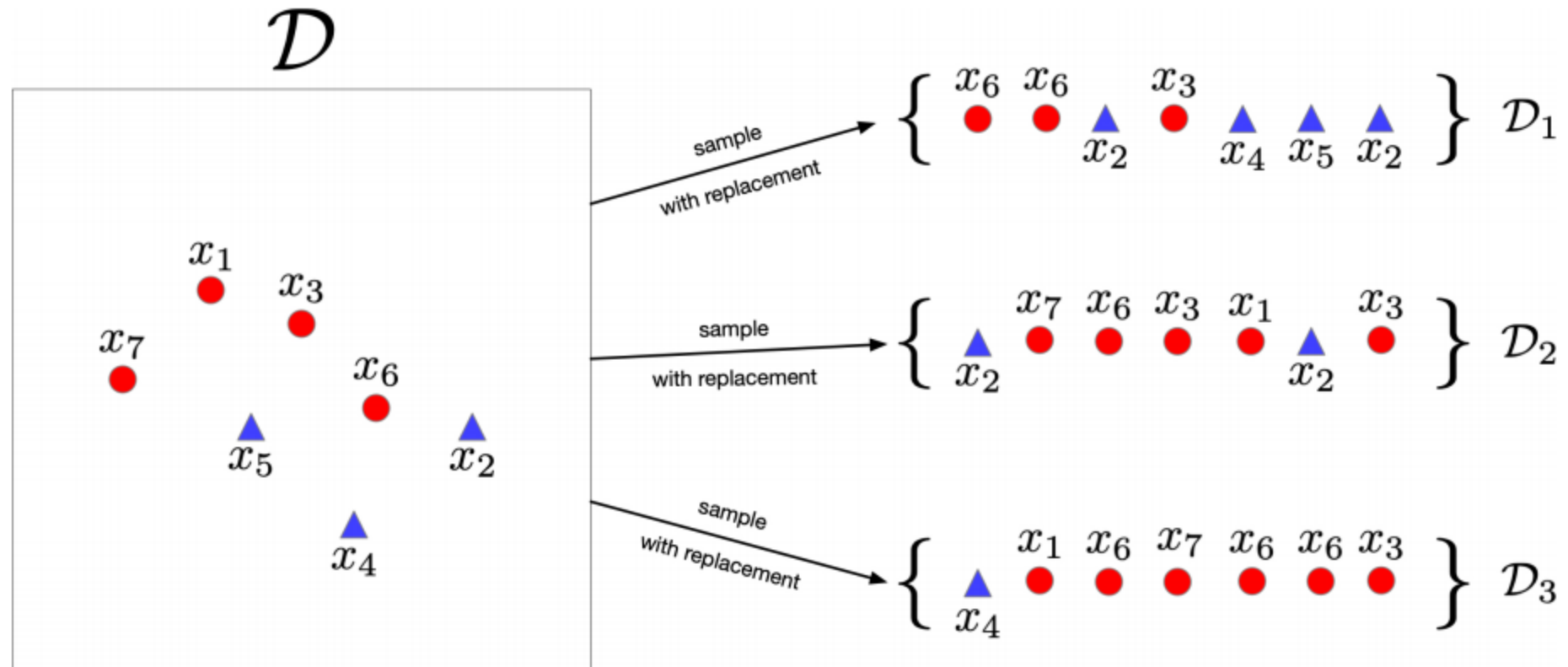


- Root entropy of class outcome: $H(Y) = -\frac{2}{7}\log_2(\frac{2}{7}) - \frac{5}{7}\log_2(\frac{5}{7}) \approx 0.86$
- Leaf conditional entropy of class outcome: $H(Y|left) \approx 0$, $H(Y|right) \approx 0.97$
- $IG(split) \approx 0.86 - (\frac{2}{7} \cdot 0 + \frac{5}{7} \cdot 0.97) \approx 0.17!$

Ensemble methods - Bagging: The Idea

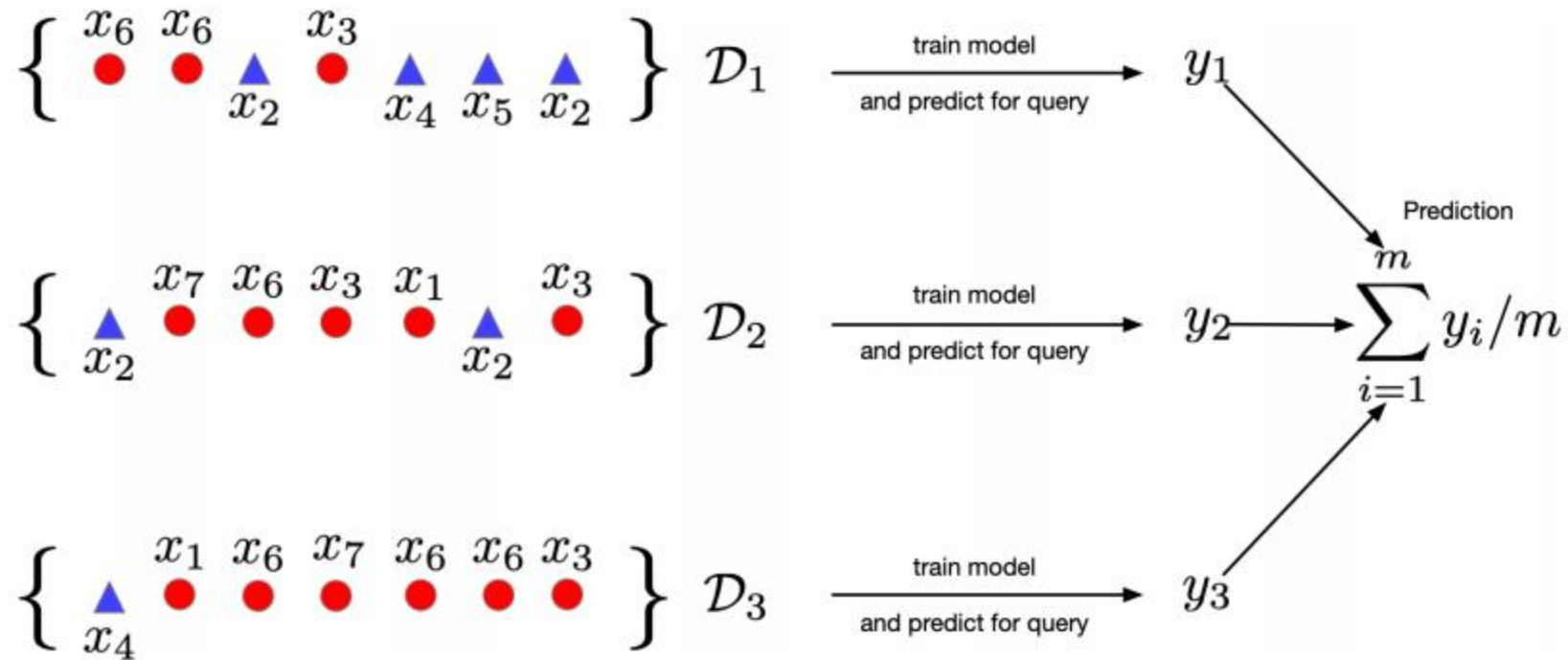
- In practice, the sampling distribution p_{sample} is often finite or expensive to sample from.
- So training separate models on independently sampled datasets is very wasteful of data!
 - Why not train a single model on the union of all sampled datasets?
- Solution: given training set $\mathcal{D}$, use the empirical distribution $p_{\mathcal{D}}$ as a proxy for p_{sample} . This is called **bootstrap aggregation**, or **bagging**.
 - Take a single dataset $\mathcal{D}$ with n examples.
 - Generate m new datasets (“resamples” or “bootstrap samples”), each by sampling n training examples from $\mathcal{D}$, with replacement.
 - Average the predictions of models trained on each of these datasets.
- The bootstrap is one of the most important ideas in all of statistics!
 - Intuition: As $|\mathcal{D}| \rightarrow \infty$, we have $p_{\mathcal{D}} \rightarrow p_{\text{sample}}$.

Bagging



In this example $n = 7, m = 3$

Bagging



Predicting on a query point x

Ensemble methods - Ada Boost Algorithm

- Input: Data $\mathcal{D}_N$, weak classifier WeakLearn (a classification procedure that returns a classifier h , e.g. best decision stump, from a set of classifiers $\mathcal{H}$, e.g., all possible decision stumps), number of iterations T
- Output: Classifier $H(x)$
- Initialize sample weights: $w^{(n)} = \frac{1}{N}$ for $n = 1, \dots, N$
- For $t = 1, \dots, T$

➤ Fit a classifier to weighted data ($h_t \leftarrow \text{WeakLearn}(\mathcal{D}_N, \mathbf{w})$), e.g.,

$$h_t \leftarrow \operatorname{argmin}_{h \in \mathcal{H}} \sum_{n=1}^N w^{(n)} \mathbb{I}\{h(\mathbf{x}^{(n)}) \neq t^{(n)}\}$$

➤ Compute weighted error $\text{err}_t = \frac{\sum_{n=1}^N w^{(n)} \mathbb{I}\{h_t(x^{(n)}) \neq t^{(n)}\}}{\sum_{n=1}^N w^{(n)}}$

➤ Compute classifier coefficient $\alpha_t = \frac{1}{2} \log \frac{1 - \text{err}_t}{\text{err}_t} (\in (0, \infty))$

➤ Update data weights

$$w^{(n)} \leftarrow w^{(n)} \exp \left(-\alpha_t t^{(n)} h_t(\mathbf{x}^{(n)}) \right) \left[\equiv w^{(n)} \exp \left(2\alpha_t \mathbb{I}\{h_t(\mathbf{x}^{(n)}) \neq t^{(n)}\} \right) \right]$$

- Return $H(\mathbf{x}) = \text{sign} \left(\sum_{t=1}^T \alpha_t h_t(\mathbf{x}) \right)$

Bagging & Boosting

- Ensembles combine classifiers to improve performance
- Boosting
 - Reduces bias
 - Increases variance (large ensemble can cause overfitting)
 - Sequential
 - High dependency between ensemble elements
- Bagging
 - Reduces variance (large ensemble can't cause overfitting)
 - Bias is not changed (much)
 - Parallel
 - Want to minimize correlation between ensemble elements.

Maximum Likelihood Estimation

- Notice, in the coin example we are actually minimizing cross-entropies!

$$\begin{aligned}\hat{\theta}_{\text{ML}} &= \max_{\theta \in [0,1]} \ell(\theta) \\ &= \min_{\theta \in [0,1]} -\ell(\theta) \\ &= \min_{\theta \in [0,1]} \sum_{i=1}^N -x_i \log \theta - (1 - x_i) \log(1 - \theta)\end{aligned}$$

- This is an example of **maximum likelihood estimation**.
 - define a model that assigns a probability (or has a probability density at) to a dataset
 - maximize the likelihood (or minimize the neg. log-likelihood).
- Many examples we've considered fall in this framework! Let's consider classification again.

Naive Bayes

- Naive assumption: **Naive Bayes** assumes that the word features x_i are **conditionally independent** given the class c .
 - This means x_i and x_j are independent under the conditional distribution $p(\mathbf{x}|c)$.
 - Note: this doesn't mean they're independent.
 - Mathematically,

$$p(c, x_1, \dots, x_D) = p(c)p(x_1|c) \cdots p(x_D|c).$$

- Compact representation of the joint distribution
 - Prior probability of class: $p(c = 1) = \pi$ (e.g. spam email)
 - Conditional probability of word feature given class: $p(x_j = 1|c) = \theta_{jc}$ (e.g. word "price" appearing in spam)
 - $2D + 1$ parameters total (before $2^{D+1} - 1$)

Naive Bayes: Learning

- The parameters can be learned efficiently because the log-likelihood decomposes into independent terms for each feature.

$$\begin{aligned}
 \ell(\boldsymbol{\theta}) &= \sum_{i=1}^N \log p(c^{(i)}, \mathbf{x}^{(i)}) = \sum_{i=1}^N \log \left\{ p(\mathbf{x}^{(i)} | c^{(i)}) p(c^{(i)}) \right\} \\
 &= \sum_{i=1}^N \log \left\{ p(c^{(i)}) \prod_{j=1}^D p(x_j^{(i)} | c^{(i)}) \right\} \\
 &= \sum_{i=1}^N \left[\log p(c^{(i)}) + \sum_{j=1}^D \log p(x_j^{(i)} | c^{(i)}) \right] \\
 &= \underbrace{\sum_{i=1}^N \log p(c^{(i)})}_{\text{Bernoulli log-likelihood of labels}} + \sum_{j=1}^D \underbrace{\sum_{i=1}^N \log p(x_j^{(i)} | c^{(i)})}_{\text{Bernoulli log-likelihood for feature } x_j}
 \end{aligned}$$

- Each of these log-likelihood terms depends on different sets of parameters, so they can be optimized independently.

Naive Bayes: Learning

- We can handle these terms separately. For the prior we maximize: $\sum_{i=1}^N \log p(c^{(i)})$
- This is a minor variant of our coin flip example. Let $p(c^{(i)} = 1) = \pi$. Note $p(c^{(i)}) = \pi^{c^{(i)}} (1 - \pi)^{1-c^{(i)}}$
- Log-likelihood:

$$\sum_{i=1}^N \log p(c^{(i)}) = \sum_{i=1}^N c^{(i)} \log \pi + \sum_{i=1}^N (1 - c^{(i)}) \log(1 - \pi)$$

- Obtain MLEs by setting derivatives to zero:

$$\hat{\pi} = \frac{\sum_i \mathbb{I}[c^{(i)} = 1]}{N} = \frac{\# \text{ spams in dataset}}{\text{total } \# \text{ samples}}$$

Naive Bayes: Learning

- Each θ_{jc} 's can be treated separately: maximize $\sum_{i=1}^N \log p(x_j^{(i)} | c^{(i)})$
- This is (again) a minor variant of our coin flip example. Let $\theta_{jc} = p(x_j^{(i)} = 1 | c)$. Note

$$p(x_j^{(i)} | c) = \theta_{jc}^{x_j^{(i)}} (1 - \theta_{jc})^{1-x_j^{(i)}}.$$

- Log-likelihood:

$$\begin{aligned} \sum_{i=1}^N \log p(x_j^{(i)} | c^{(i)}) &= \sum_{i=1}^N c^{(i)} \left\{ x_j^{(i)} \log \theta_{j1} + (1 - x_j^{(i)}) \log(1 - \theta_{j1}) \right\} \\ &\quad + \sum_{i=1}^N (1 - c^{(i)}) \left\{ x_j^{(i)} \log \theta_{j0} + (1 - x_j^{(i)}) \log(1 - \theta_{j0}) \right\} \end{aligned}$$

- Obtain MLEs by setting derivatives to zero:

$$\hat{\theta}_{jc} = \frac{\sum_i \mathbb{I}[x_j^{(i)} = 1 \ \& \ c^{(i)} = c]}{\sum_i \mathbb{I}[c^{(i)} = c]} \quad \text{for } \underline{c=1} \quad \frac{\text{\#word } j \text{ appears in spams}}{\text{\# spams in dataset}}$$

Naive Bayes: Inference

- We predict the category by performing **inference** in the model.
- Apply **Bayes' Rule**:

$$p(c | \mathbf{x}) = \frac{p(c)p(\mathbf{x} | c)}{\sum_{c'} p(c')p(\mathbf{x} | c')} = \frac{p(c) \prod_{j=1}^D p(x_j | c)}{\sum_{c'} p(c') \prod_{j=1}^D p(x_j | c')}$$

- We need not compute the denominator if we're simply trying to determine the most likely c .
- Shorthand notation:

$$p(c | \mathbf{x}) \propto p(c) \prod_{j=1}^D p(x_j | c)$$

- For input $\mathbf{x}$, predict by comparing the values of $p(c) \prod_{j=1}^D p(x_j | c)$ for different c (e.g. choose the largest).

Naive Bayes

- Naive Bayes is an amazingly cheap learning algorithm!
- **Training time**: estimate parameters using maximum likelihood
 - Compute co-occurrence counts of each feature with the labels.
 - Requires only one pass through the data!
- **Test time**: apply Bayes' Rule
 - Cheap because of the model structure. (For more general models, Bayesian inference can be very expensive and/or complicated.)
- We covered the Bernoulli case for simplicity. But our analysis easily extends to other probability distributions.
- Unfortunately, it's usually less accurate in practice compared to discriminative models due to its "naive" independence assumption.

Gaussian Mixture Model (GMM)

→ 假设每一个数据点是从K个聚类中选一个

What is $p(x)$? 高斯混合模型边缘函数的计算公式

选聚类概率率是 π_k

再从该聚类对应高斯分布

采样得到

$$p(\mathbf{x}) = \sum_{k=1}^K p(z = k)p(\mathbf{x}|z = k) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \mathbf{I})$$

$$\sum_{k=1}^K \pi_k = 1$$

- This distribution is an example of a **Gaussian Mixture Model (GMM)**, and π_k are known as the **mixing coefficients**
- In general, we would have different covariance for each cluster, i.e., $p(\mathbf{x}|z = k) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$. For this lecture, we assume $\boldsymbol{\Sigma}_k = \mathbf{I}$ for simplicity.

EM Algorithm for GMM

目的优化 $\hat{\mu}_k, \hat{\pi}_k$ 从 $p(\mathbf{x})$ 中, 让 GMM 的 $p(\mathbf{x})$ 逼近真实分布

- Initialize the means $\hat{\mu}_k$ and mixing coefficients $\hat{\pi}_k$
- Iterate until convergence:

➤ **E-step:** Evaluate the responsibilities $r_k^{(n)}$ given current parameters

$$r_k^{(n)} = p(z^{(n)} = k | \mathbf{x}^{(n)}) = \frac{\hat{\pi}_k \mathcal{N}(\mathbf{x}^{(n)} | \hat{\mu}_k, \mathbf{I})}{\sum_{j=1}^K \hat{\pi}_j \mathcal{N}(\mathbf{x}^{(n)} | \hat{\mu}_j, \mathbf{I})} = \frac{\hat{\pi}_k \exp\{-\frac{1}{2} \|\mathbf{x}^{(n)} - \hat{\mu}_k\|^2\}}{\sum_{j=1}^K \hat{\pi}_j \exp\{-\frac{1}{2} \|\mathbf{x}^{(n)} - \hat{\mu}_j\|^2\}}$$

数据点 贝叶斯

➤ **M-step:** Re-estimate the parameters given current responsibilities

$$\hat{\mu}_k = \frac{1}{N_k} \sum_{n=1}^N r_k^{(n)} \mathbf{x}^{(n)} \quad (\text{求导})$$

$$\hat{\pi}_k = \frac{N_k}{N} \quad \text{with} \quad N_k = \sum_{n=1}^N r_k^{(n)} \quad (\text{拉格朗日})$$

➤ Evaluate log likelihood and check for convergence

$$\log p(\mathcal{D}) = \sum_{n=1}^N \log \left(\sum_{k=1}^K \hat{\pi}_k \mathcal{N}(\mathbf{x}^{(n)} | \hat{\mu}_k, \mathbf{I}) \right)$$

Relation to k-Means

- The K-Means Algorithm:
 1. **Assignment step**: Assign each data point to the closest cluster
 2. **Refitting step**: Move each cluster center to the average of the data assigned to it
- The EM Algorithm:
 1. **E-step**: Compute the posterior probability over z given our current model
 2. **M-step**: Maximize the probability that it would generate the data it is currently responsible for.